MILITARY SPECIFICATION

SIGHT, HELICOPTER, HEAD-UP SUBSYSTEM: M76

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification contains quality assurance provisions and preparation for delivery for the Sight, Helicopter, Head-Up Subsystem (HUDS): M76.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems
MIL-I-8500	Interchangeability and Replaceability of Components Parts for Aircraft and Missiles
MIL-Q-9858	Quality Control Requirements
MIL-F-13926	Fire Control Material: General Specification Governing the Manufacturer and Inspection of

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MIL-O-13830	Optical Component for Fire Control Instrument, General Specification Governing Assembly and Inspection of
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral Aircraft Instrument, General Specification for
MIL-M-38510	Microcircuits, General Specification for
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for
MIL-S-46844	Solder Bath, Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-W-63150	Weapons and Support Material, Standard Quality Assurance Provisions for
MIL-D-70443	Display Unit, Sight, Helicopter Head Up Subsystem: M76
MIL-P-70444	Processor, Signal, Sight, Helicopter Head Up Subsystem: M76

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

11780000	Head Up Display Subsystem
11780001	Head Up Display
11780002	Signal Processor
11780246	Mount, HUD Boresightable

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein:

American Society for Testing and Materials (ASTM).

ASTM - D3951 - Standard Practice for Commercial Packaging.

3. REQUIREMENTS.

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications, and standards.

3.3 Head Up Display Subsystem (HUDS). The HUDS shall comply with all requirements specified on Drawing (Dwg) 11780000, all associated drawings, and with all requirements specified in applicable specifications and standards. The HUDS covered by this specification shall be configured for use on the AH-1S series helicopters. The HUDS shall provide a visual presentation of flight and fire control data to the pilot during day and night missions. The data shall appear on the Head Up Display (HUD). The HUD shall provide weapon aiming in hover and forward flight and shall enable the pilot to align the helicopter for launching TOW missiles and to fly at low altitudes without scanning the instrument panel for critical flight information.

3.3.1 Major components list. Major components of the HUDS are as follows:

Nomenclature	Drawing
Head Up Display	11780001
Signal Processor	11780002
Mount, HUD Boresightable	11780246

3.4 Characteristics.

3.4.1 Electrical.

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3.4.1.1 Electrical signal interface. The HUDS signal interface with the AH-1 series helicopter shall be as defined in Appendix A.

3.4.1.2 Electrical power.

3.4.1.2.1 Steady state. The electrical service conditions will be in accordance with MIL-STD-704, Category "B" utilization equipment, except that the transient frequency limit of Figure 5, limit 4 in MIL-STD-704 shall be 240 Hz. The equipment shall meet all specified requirements of MIL-STD-704 for the types of power specified herein, and shall give specified performance when supplied with electrical power having these characteristics and limits. The following power inputs will be available to the HUDS:

a. 28 Vdc for HUDS primary operating voltage, maximum power available shall be 76 watts for test and 70 watts for normal operation.

b. 0 to 28 Vdc instrument panel dimmable lighting bus; maximum power available for this function shall be 5 watts.

c. 115 Vac 400 Hz single phase for SP cooling fan and HUDS elapsed time indicators; maximum available power shall be 15 volt-amperes.

3.4.1.2.2 Electrical power interruptions. Momentary power interruptions to the HUDS equipment will be experienced when external primary power is switched to helicopter power or when switching from one helicopter power bus to another. The HUDS shall meet the performance requirements of this specification within five seconds after such momentary power interruptions as defined in MIL-STD-704. The HUDS shall return to specified performance within fifteen seconds (10 seconds desired) after any similar momentary longer term interruption.

3.4.1.2.3 Polarity reversal. Components that receive dc helicopter primary or instrument power shall not be damaged when subjected to a polarity reversal due to inadvertant polarity reversal of the helicopter dc power. HUDS operation shall return to normal after polarity is restored, within the constraints specified for electrical power interruptions specified in 3.4.1.2.2

3.4.2 Performance.

3.4.2.1 Operation. The HUDS is operated by the pilot primarily for aiming the helicopter to fire rockets and gun and to assist the copolot/gunner in operating the TOW missile system.

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The pilot also uses the HUDS to observe displayed data for flight safety. The pilot observes these data items from his normal head position. When certain fire control data are presented, requiring large translation of fire control symbols, additional head motion shall be permitted. The pilot shall have control of display operation with controls located on the HUD, and shall control displayed data by the use of externally generated command signals.

3.4.2.2 Power control modes. The HUDS shall have the following power modes:

a. Off. The HUDS shall not be operational; all electrical power shall be removed from HUDS circuits.

b. Standby. All necessary power shall be applied for HUDS operations. The Cathode Ray Tube (CRT) shall be blanked.

c. On. The HUDS shall be fully operational in this mode.

3.4.2.3 Operational modes. When the HUDS is in the "ON" power control mode, the HUDS shall have the following selectable operating modes. The data and symbols displayed within each mode shall be in accordance with Table I.

3.4.2.3.1 Stadiametric mode. Whenever the HUD is placed in the "STAD" mode, the Stadiametric Reticle (SMR) shall be displayed and its brightness controlled. The SMR shall be a backup reticle that is selected for display in the event of primary failure in the SP's interface, processor, or main program memory circuitry.

3.4.2.3.2 Normal mode. The following data and symbols shall be displayed when the HUDS is in "NORMAL" mode and the appropriate control signals are received as specified in Table I.

3.4.2.3.2.1 Boresight reference symbol. The boresight reference symbol, which serves as the HUDS display reference, shall always be displayed.

3.4.2.3.2.2 Normal data. The following alpha numeric data shall always be displayed when the data exists at the HUDS interface with the condition that the FCC operation is valid.

a. Range
b. Engine torque
c. Magnetic heading
d. Radar altitude

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Table I. Symbology Vs. Operating Mode

SYMBOLS \ OPERATING MODES	NORMAL MODE (1)							Stadia-Metric Mode	Test Mode
	Direct Rocket	Indir Rocket	Back-Up Rocket	Fixed Gun	Tow Attack	Tow Ready	Tow Fired		
Range	X	X	X (5)	X (7)	X (7)	X (7)	X (7)		X (5)
Engine Torque	X	X		X (6)	X (6)	X (6)	X (6)		X
Magnetic Heading	X	X		X (6)	X (6)	X (6)	X (6)		X
Radar Altitude	X	X		X (6)	X (6)	X (6)	X (5)		X
Fire Control Reticles (12) (2)	XA	XB	XC (5)	XA (3)					XC (5)
Copilot/Gunner Los Sym	X	X	X	X	X (4)	X	X		X (4) (12)
Boresight Ref Sym	X	X	X	X	X	X	X		X
Tow Ascend/Descend					X	X			X
Az Course Pointer/Scale (12)	X	X	X	X	X	X	X		X (12)
Tow Prelaunch Sym					X	X			X
Tow Postlaunch Sym							X		X
SMR Sym								X	
Radar Alt High Limit	X	X	X	X	X	X	X		X
Radar Alt Low Limit	X	X	X	X	X	X	X		X
Left Missile Select					X (8)	X (8)			X
Right Missile Select					X (9)	X (9)			X

X Denotes Symbol Displayed

NOTES: (1)

MODE LOGIC TABLE

MODES	Input Discretes					
	Direct/ Indirect	Gun	*FCC Good	Tow Atk	Tow Rdy	Tow Fired
Direct Rocket	ON**	OFF	ON	OFF	N/A	OFF
Indirect Rocket	OFF	OFF	ON	OFF	N/A	OFF
Back-Up Rocket	N/A***	OFF	OFF	OFF	N/A	OFF
Fixed Gun	N/A	ON	N/A	OFF	N/A	OFF
Tow Attack	N/A	N/A	N/A	ON	OFF	N/A
Tow Ready	N/A	N/A	N/A	ON	ON	N/A
Tow Fired	N/A	N/A	N/A	OFF (13)	OFF (13)	ON

* ON = FCC Valid; OFF = FCC Invalid

**ON = Direct; OFF = Indirect

***N/A = No Affect

(2) XA - FC Direct Rocket Reticle
XB - FC Indirect Rocket Reticle
XC - FC Backup Rocket Reticle

(3) Fixed Position at ADL.

(4) Flashes at 4 Hz Rate.

(5) Use LRF Data for Elevation/Depresion and Range Display.

(6) If FCC Good Discrete is not Present, Symbol is Blanked.

(7) If FCC Good Discrete is not Present, Laser Range Displayed.

(8) Left Missile Select Discrete is Present.

(9) Right Missile Select Discrete is Present.

(10) N/A

(11) N/A

(12) Caution Symbol is Overlayed on Reticles/ Pointers if Limits are Reached or Exceeded.

(13) When all Discretes, Tow Atk, Tow Rdy, and Tow Fired, are on, an X shall be placed over the selected missile in the prelaunch constraint symbology. All three signals will be in the ON condition for 1.52 seconds during a tow missile launch sequence.

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If the FCC operation is invalid, the raw laser range shall be displayed and so indicated. The Radar Altitude High/Low symbols shall be displayed when these discrete commands from the radar altimeter exists at the HUDS interface.

3.4.2.3.2.3 Fire control symbol. A fire control reticle (see Figures 1 and 2) used for either direct for indirect rocket fire shall be displayed when the HUDS is in the appropriate weapon mode as shown in Table I. The reticle shall be capable of being displaced to any location within the HUD field of view (see Figure 3) as a function of azimuth and elevation position command signals from the FCC. When a gun mode discrete is present, the fire control direct rocket reticle shall be superimposed on the boresight reference symbol by the HUDS. If the FCC has failed and gun mode has not been selected, the fire control back-up rocket reticle (see Figure 2) shall be positioned in elevation as a function of raw laser range by the HUDS.

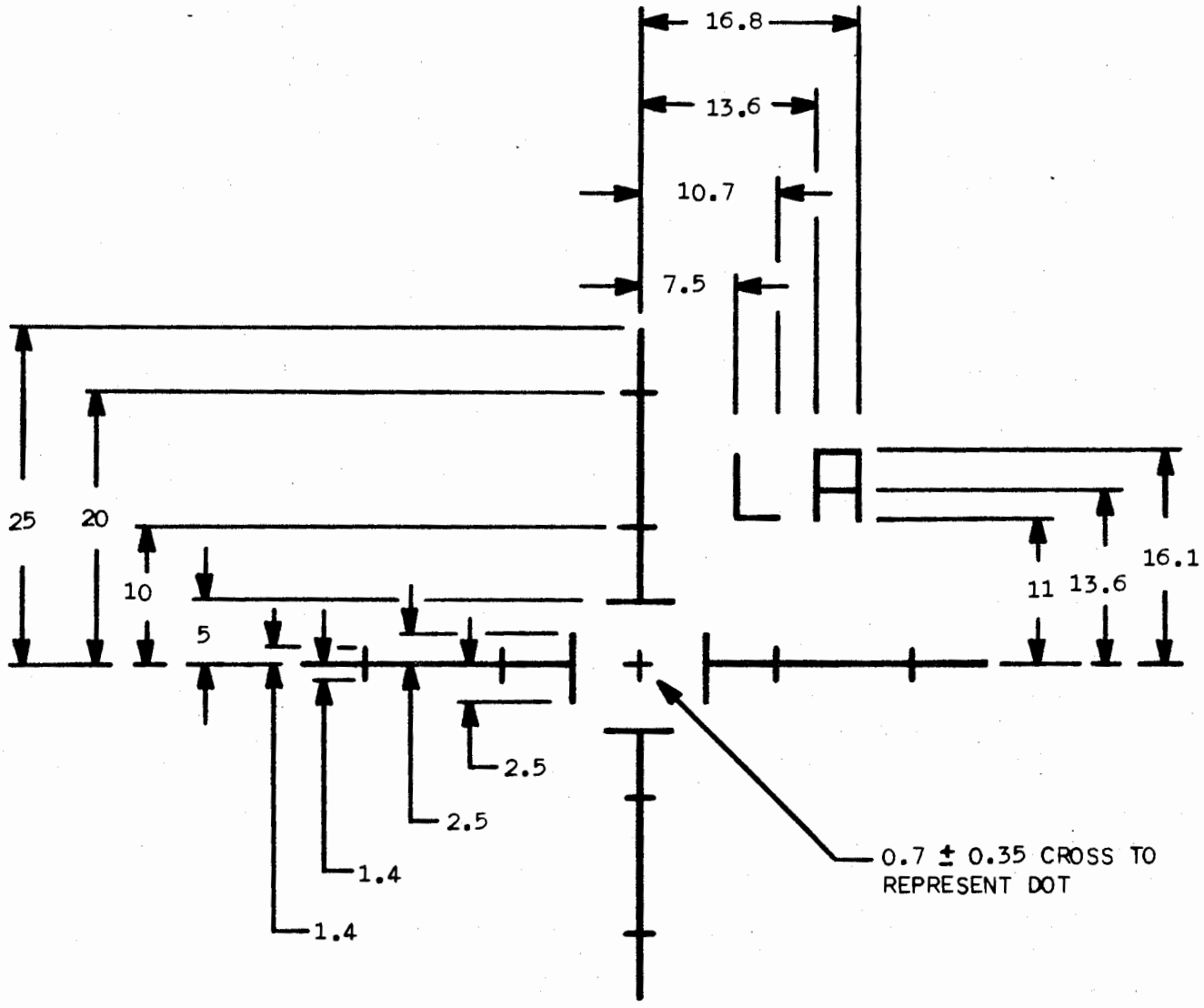
3.4.2.3.2.4 TOW mode symbols. Pilot steering information symbols, pre-launch or post-launch, shall be displayed, when the weapon system is in the TOW mode and when the HUDS receives the appropriate TOW mode discrete signals. The two sets of symbols shall be functionally identical but shall be represented in a different scale factor for each of the two modes. Symbols indicating right or left missile selection shall be displayed whenever the appropriate discrete signal occurs.

3.4.2.3.2.5 TSU LOS cueing symbol. A TSU LOS cueing symbol shall be variable in location in the field of view as a function of azimuth and elevation position and mode commands to the HUDS.

3.4.2.3.2.6 TSU coarse azimuth symbols. A coarse azimuth pointer and scale representing the total azimuth travel of the TSU shall be displayed.

3.4.2.3.3 Test mode. All data positions and symbols shall be displayed simultaneously as shown in Figure 4.

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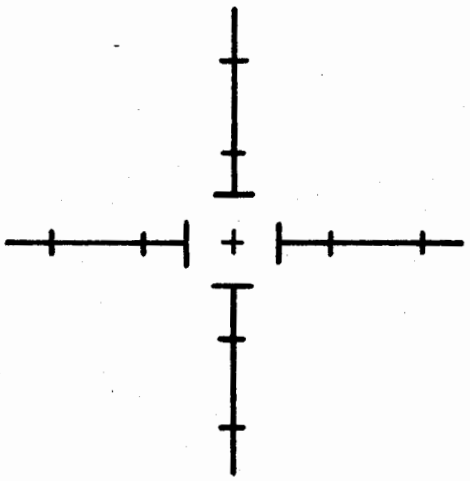


NOTE: (1) All dimensions are in milliradians (mr).

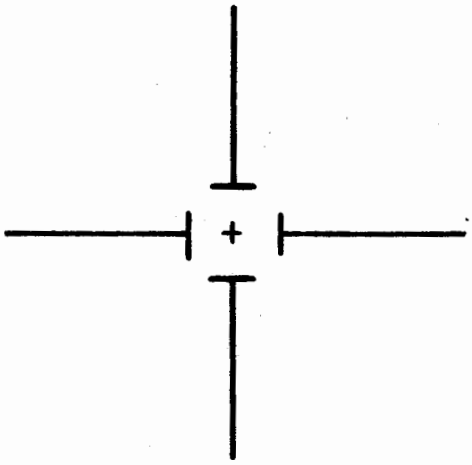
(2) Unless otherwise specified, all dimensions shall have a ± 1.0 mr tolerance.

Figure 1. Fire Control Reticle

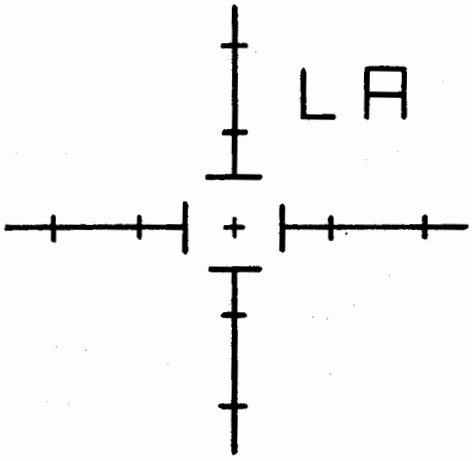
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DIRECT ROCKET RETICLE
(*XA)



INDIRECT ROCKET RETICLE
(*XB)

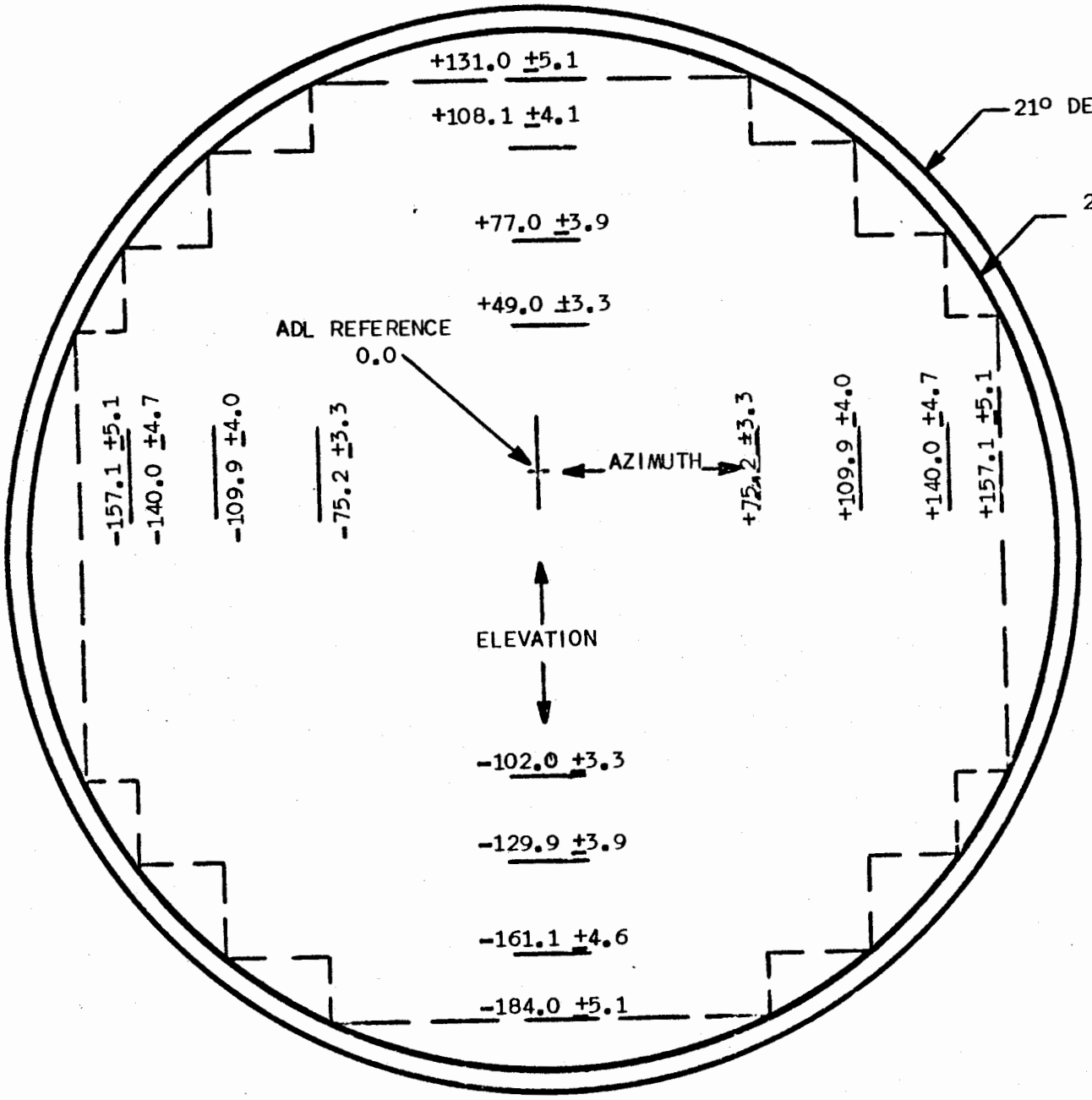


BACK-UP DIRECT ROCKET RETICLE
(*XC)

*REFER TO TABLE II

Figure 2. Fire Control Reticle Formats

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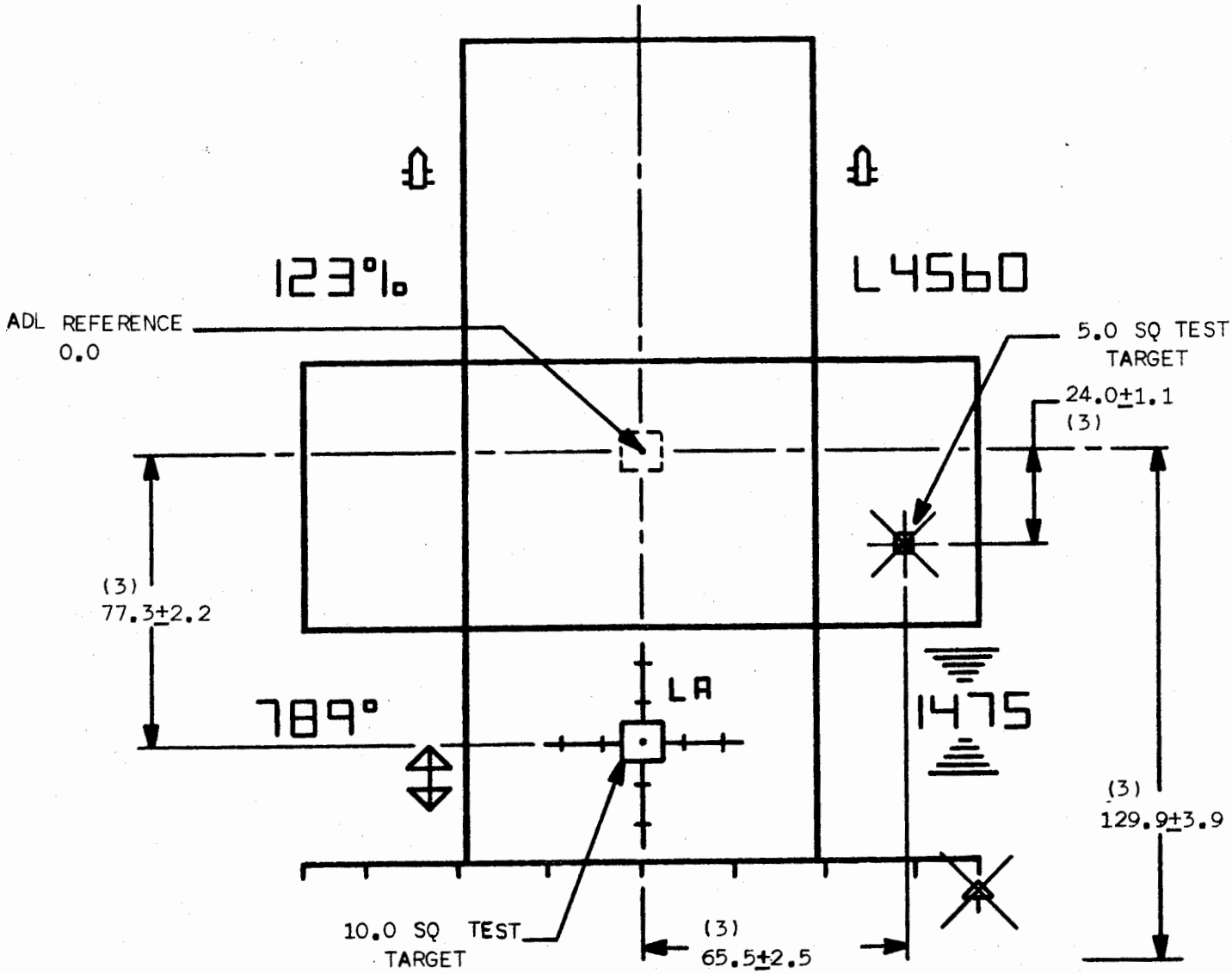
NOTE:

(1) FIELD OF VIEW LIMITS FOR THE CENTER OF THE FIRE CONTROL RETICLE AND TSU LOS CUEING RETICLE WHEN NOT IN TOW MODE.

(2) ALL DIMENSIONS ARE IN MILLIRADIANS.

FIGURE 3. FIELD OF VIEW POSITIONING LIMITS

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NOTES:

(1) ALL DIMENSIONS ARE IN MILLIRADIANS (mr).

(2) UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS SHALL HAVE A ±1.0 mr TOLERANCE.

(3) TEST TARGET AZ OR ELV LOCATION ONLY.

FIGURE 4. TEST MODE

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3.4.2.4 Data and symbol characteristics.

3.4.2.4.1 SMR characteristics. The SMR shown in Figure 5 shall provide a boresight check and auxillary sighting symbol for the central portion of the pilot's field of view.

3.4.2.4.2 Boresight reference symbol. The boresight reference symbol (see Figure 6) will serve as the ADL reference symbol during flight.

3.4.2.4.3 Numeric data. The following data shall be presented in numeric form on the HUD (see Figure 7). When either the center of the fire control reticle or the center of the gunner's LOS cueing reticle is positioned within the limits around the numeric data (see Figure 7), the corresponding numeric data shall be blanked.

a. Range. A four digit number, with last digit always a zero, and a source indicator: L for raw laser, F for FCC smoothed range.

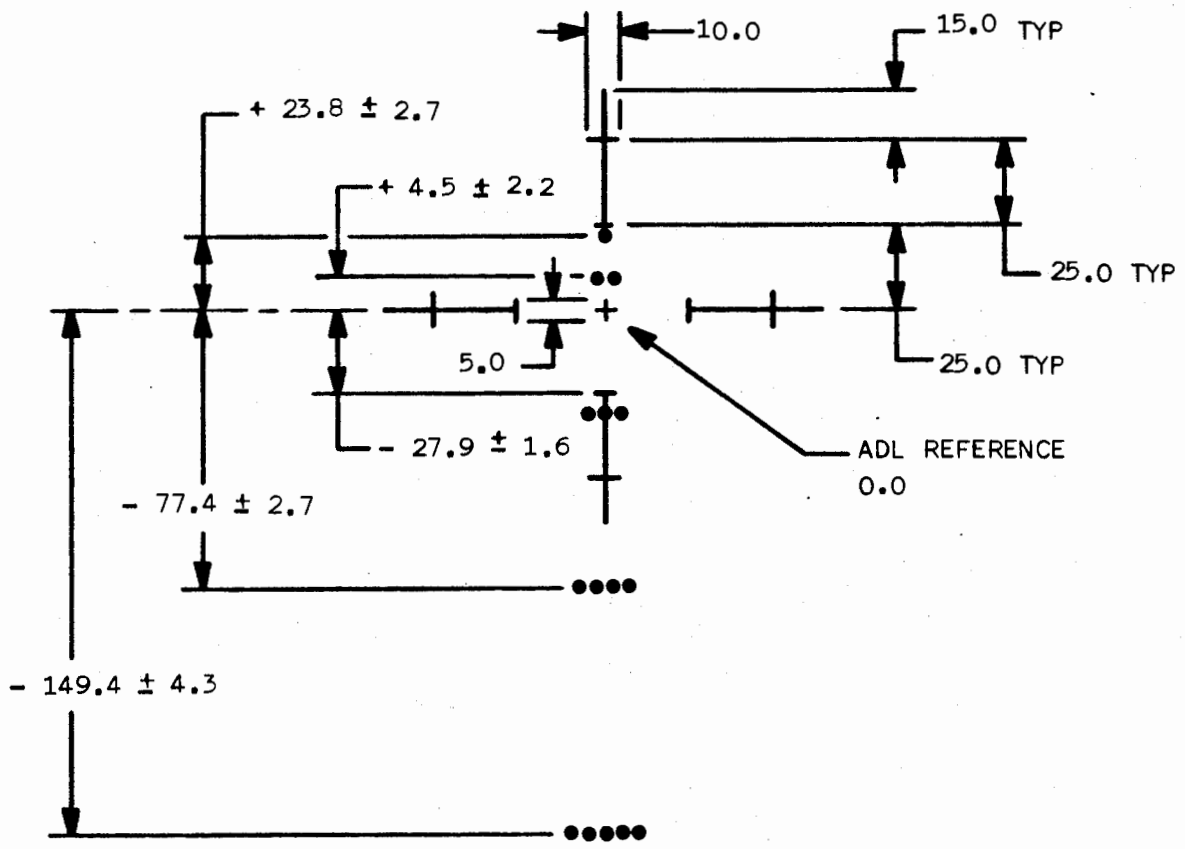
b. Engine torque. A three digit number followed by a percent sign.

c. Magnetic heading. A three digit number followed by a degree sign. Antecedent zeros shall be displayed.

d. Radar altitude. A four digit number, displaying radar altitude in increments of 5 feet. This number shall also have an upper and lower "limit exceeded" indicator over and under the displayed number, respectively (see Figure 4). When the radar altitude input data exceeds 1499 feet, the numerics shall be blanked and not unblanked until the input data drops below 1475 feet.

3.4.2.4.4 Fire Control reticle. The fire control reticle shall be configured as shown in Figures 1 and 2. The reticle shall also be variable in azimuth and/or elevation to any location within the field-of-view limits shown in Figure 3. A caution symbol (see Figure 8) shall be overlayed on the reticle when limit is reached.

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Notes:

(1) All dimensions are in milliradians (mr).

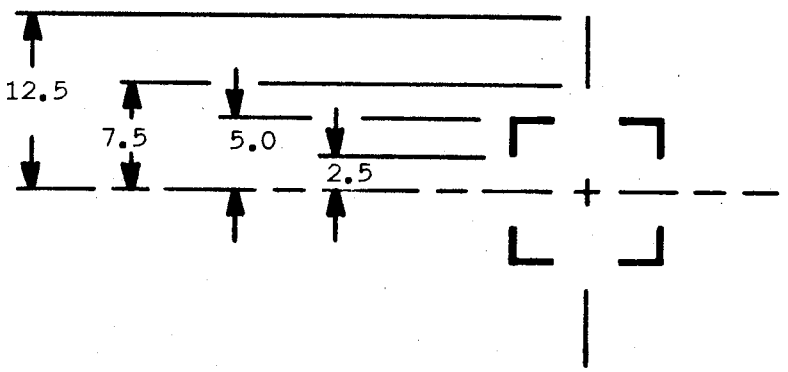
(2) Spacing between dots is 5.0 mr.

(3) ADL cross is 5.0 mr.

(4) Unless otherwise specified, all dimensions shall have a ± 1.0 mr tolerance.

Figure 5. STADIAMETRIC RETICLE

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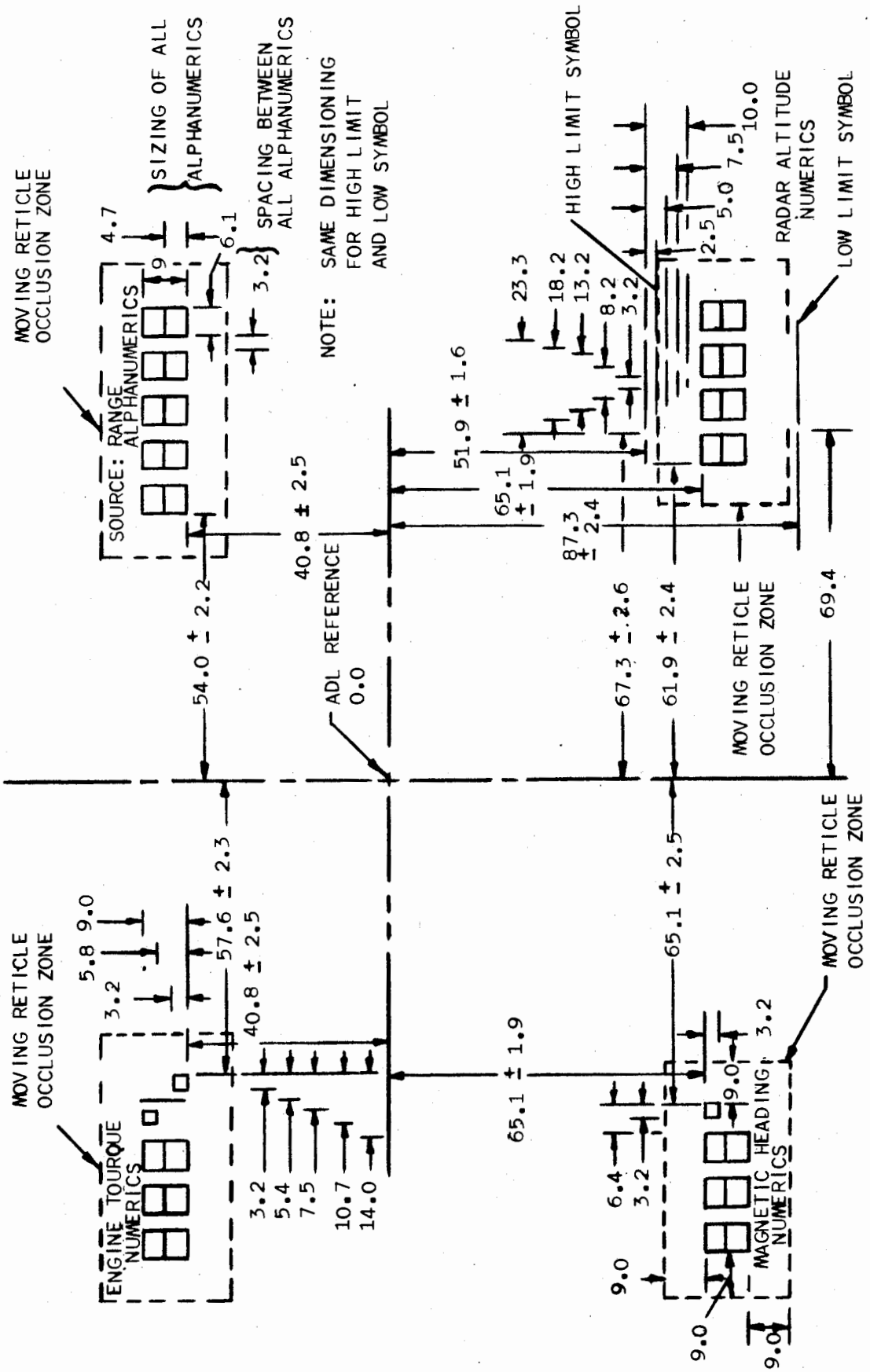


NOTE: (1) All Dimensions are in milliradians (mr)

(2) ALL dimensions shall have a ±1.0 mr tolerance.

Figure 6. Boresight Reference Symbol

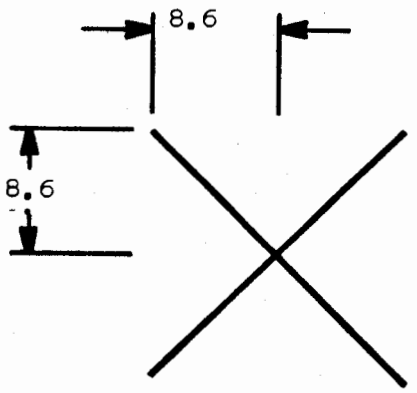
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NOTE: (1) ALL DIMENSIONS ARE IN MILLIRADIANS (mr).
(2) UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS SHALL HAVE A ± 1.0 mr TOLERANCE.

FIGURE 7. FLIGHT DATA NUMERICS

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NOTE: (1) ALL dimensions are in milliradians (mr).

(2) ALL dimensions shall have a 1.0 mr tolerance.

Figure 8. Caution Symbol

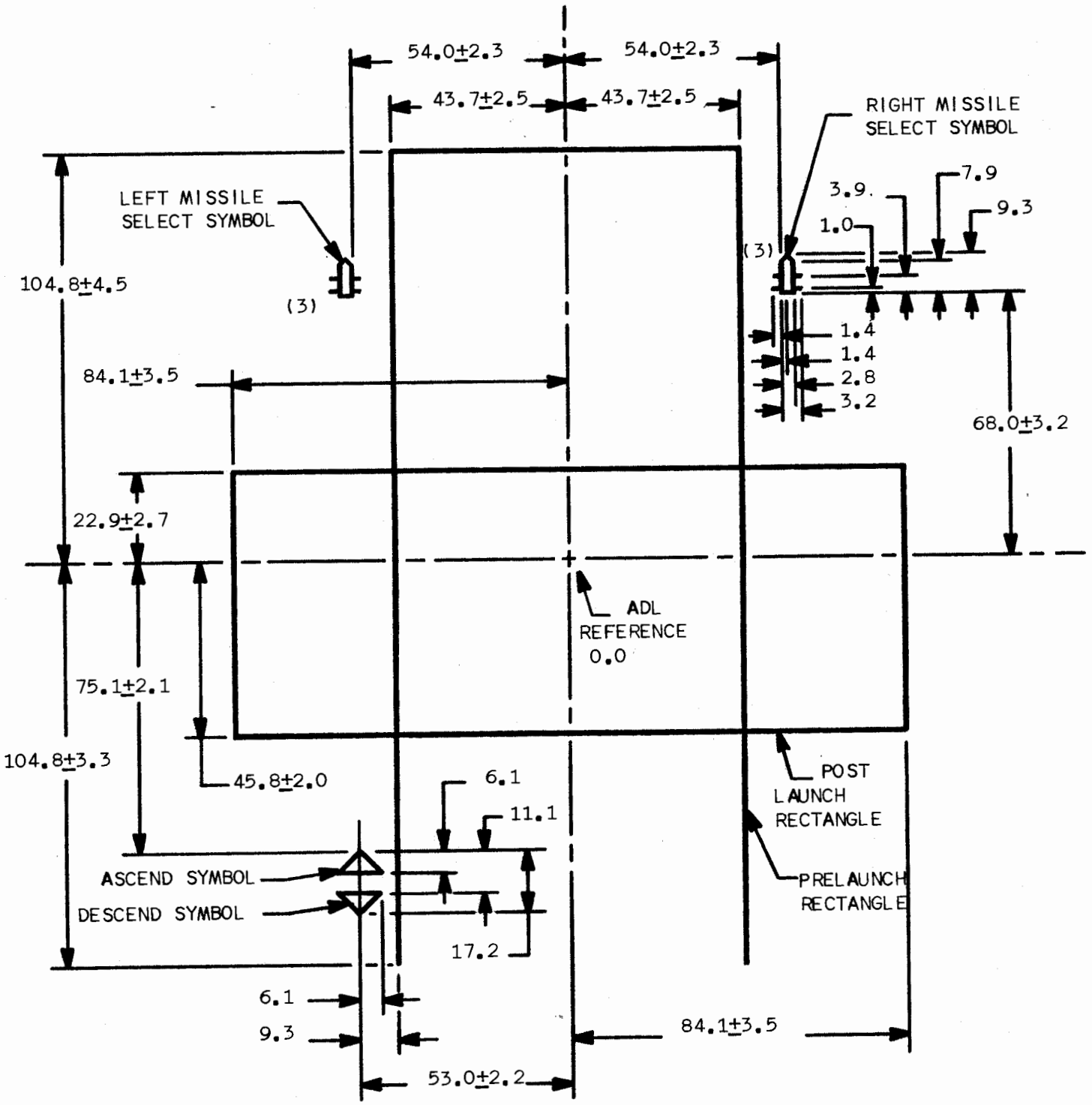
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3.4.2.4.5 TOW mode symbols. The TOW mode symbols shall be configured as shown in Figure 9 and displayed as shown in Table I. The prelaunch rectangle shall represent launch limits superimposed 1:1 over the real world before launch of the TOW missile. After TOW launch, the post launch rectangle shall be scaled to represent gunner's sight azimuth and elevation maneuver limits, at a scale factor of approximately 23:1 in azimuth and in elevation. A stationary symbol pair representing ascend and descend (nose up/nose down) commands shall also be displayed in the prelaunch mode. The left and right TOW missile select symbol shall be displayed when commanded by the Left Missile Select discrete and the Right Missile discrete respectively.

3.4.2.4.6 Gunner's TSU LOS cueing symbol. A Gunner's TSU LOS cueing symbol shall be displayed as shown in Figure 10. The primary azimuth and elevation positioning commands shall be from the FCC. When the FCC good discrete is false, the HUDS shall utilize position commands from the TSU. In the TOW prelaunch mode, the LOS cueing symbol's commanded position shall be scaled 1:1 over the area defined by $\pm$ 3.75 degrees in azimuth and +7.5 to -10.5 degrees in elevation with respect to the ADL as shown in Figure 11. When in the TOW post launch mode, the LOS cueing symbols commanded position shall be linearly scaled so as to always remain within the area shown in Figure 12. When the LOS cueing symbol reaches the angular limits shown in Figure 11 and 12, the caution symbol shown in Figure 8 shall be superimposed on the LOS cueing symbol. When not in TOW mode, the LOS cueing symbol's commanded position shall be scaled 1:1 with respect to the real world over the area shown in Figure 3. When the LOS cueing symbol reaches the limits, the caution symbol shall be superimposed on the LOS cueing symbol and shall remain there until the actual LOS commanded position is within the limits. When in the TOW Attack Mode, the LOS cueing reticle shall be flashed by the HUD at a rate of 4 $\pm$ 1 times per second with a 50 $\pm$ 20 percent duty cycle.

3.4.2.4.7 TSU coarse azimuth symbols. An azimuth scale and pointer which linearily represents $\pm$ 110 degree total azimuth travel of the TSU shall be displayed as shown in Figure 10. The scale shall have markers represent 1, $\pm$ 30, $\pm$ 60, $\pm$ 90, and $\pm$ 110 degrees azimuth angles. The caution symbol shall be superimposed on pointer when the TSU coarse azimuth input is $\pm$ 110 degrees. The primary azimuth positioning command shall be from the TSU.

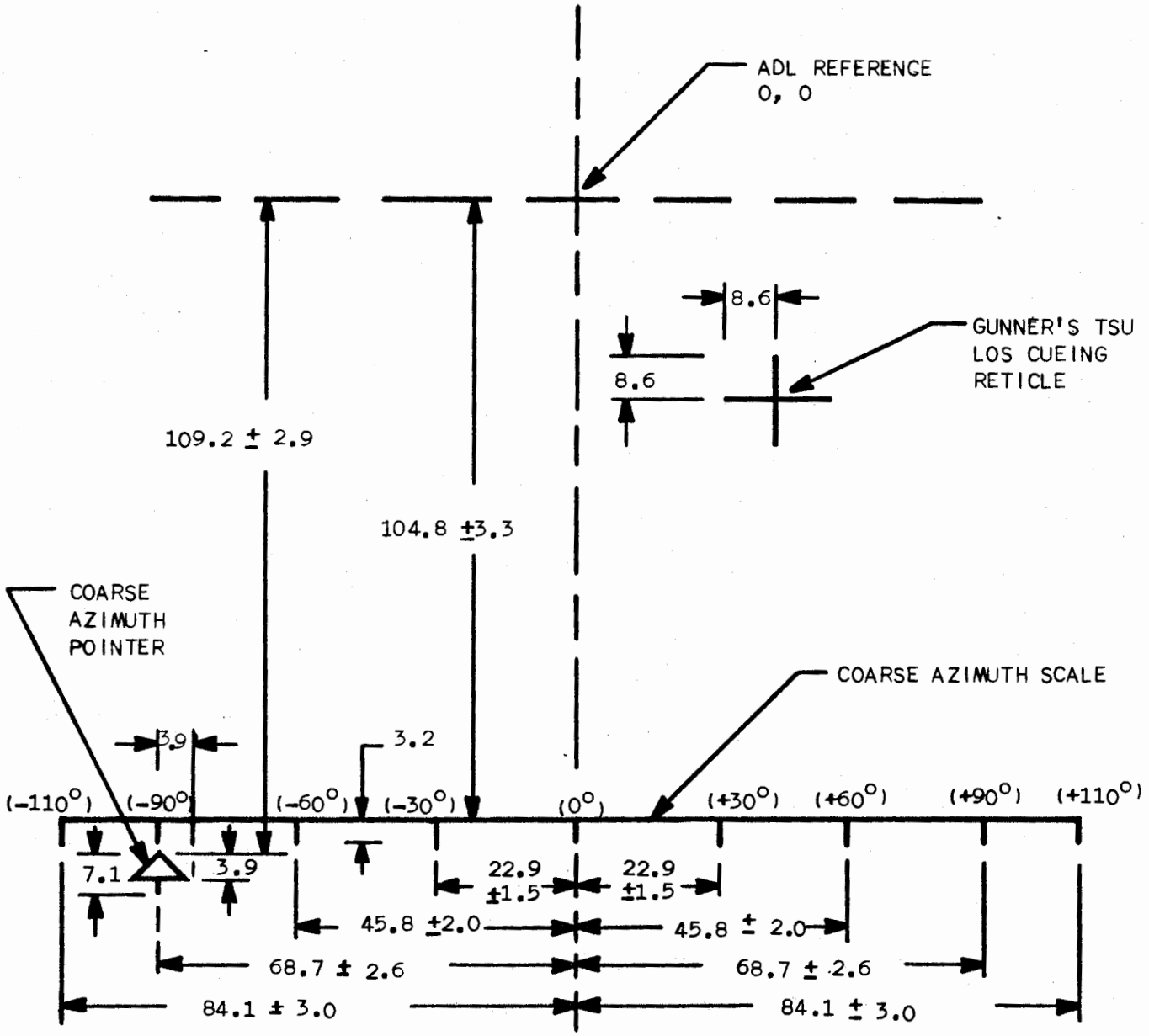
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NOTES:
(1) ALL DIMENSIONS ARE IN MILLIRADIANS (mr).
(2) UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS SHALL HAVE A ±1.0 mr TOLERANCE.
(3) AN X WILL APPEAR OVER MISSILE IMAGE WHEN TRIGGER IS PULLED.

FIGURE 9. TOW MODE SYMBOLOGY

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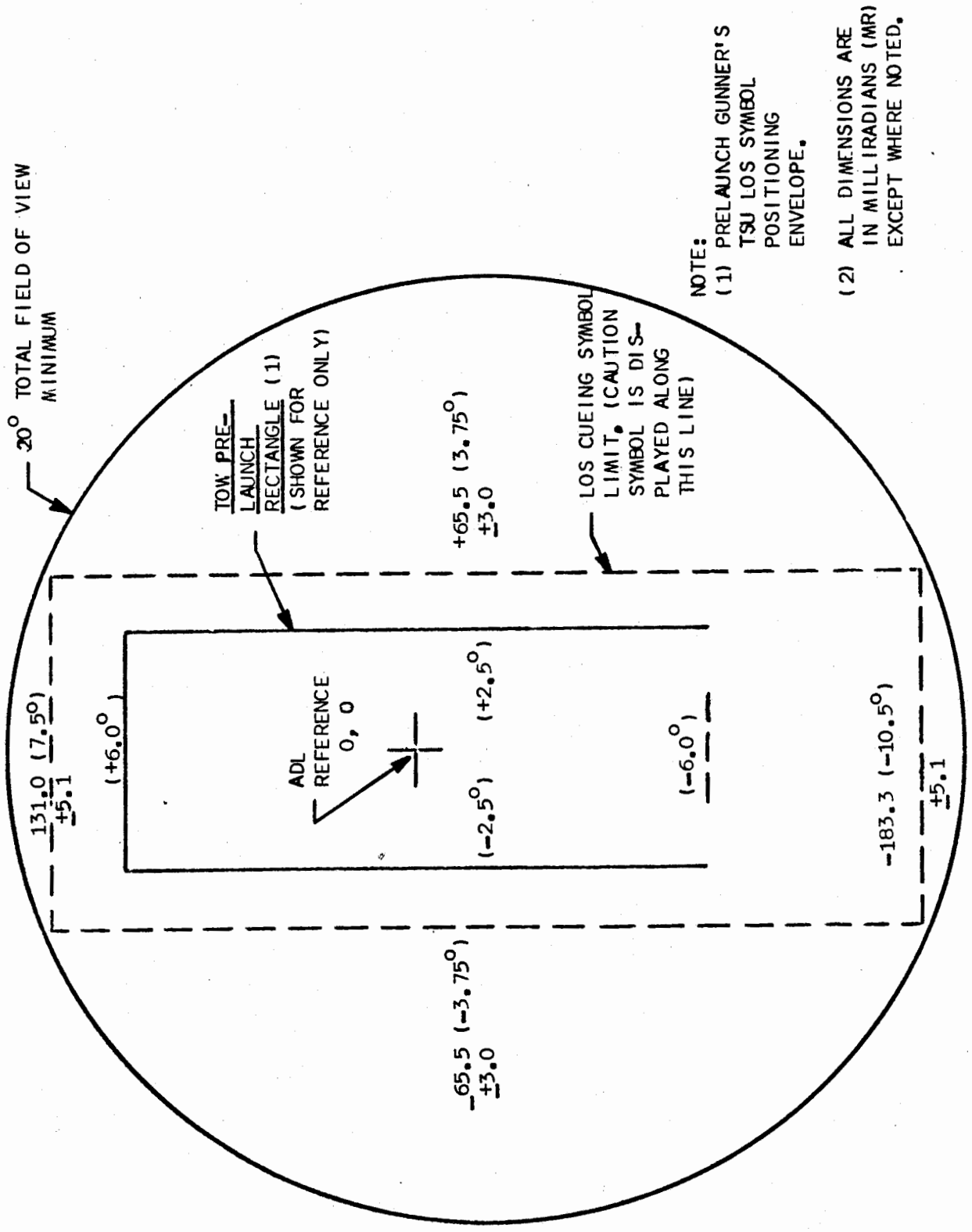
Notes: (1) All dimensions are in milliradians (mr).

(2) Unless otherwise specified, all dimensions shall have a ± 1.0 mr tolerance.

Figure 10. TSU SYMBOLOGY

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FIGURE 11. TOW PRELAUNCH TSU LOS RETICLE LIMITS



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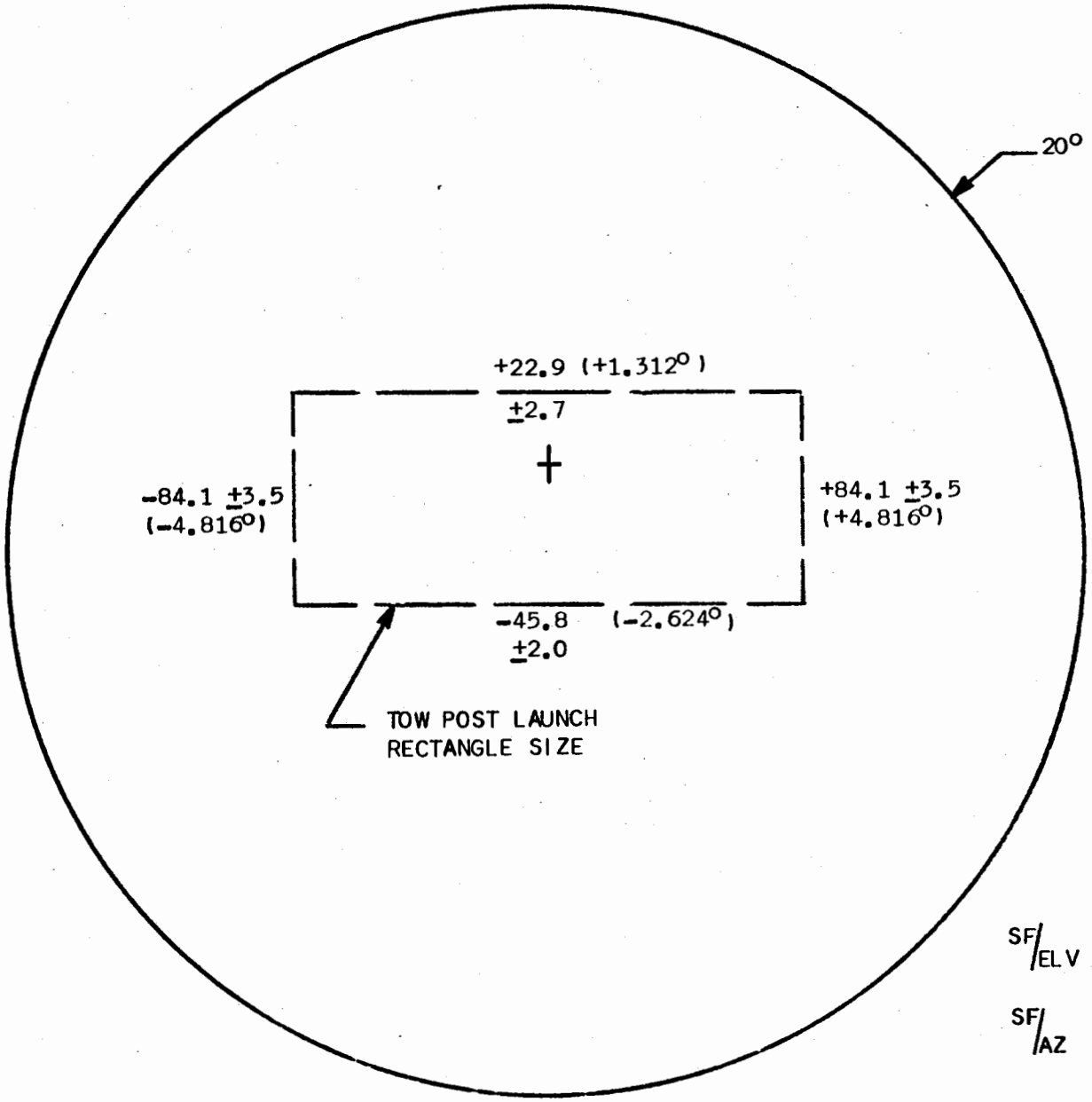


FIGURE 12. TOW POST LAUNCH TSU LOS RETICLE

NOTE:
(1) POST LAUNCH GUNNER'S TSU LOS SYMBOL POSITIONING ENVELOPE.

$$\text{SCALE FACTOR (SF)} = \frac{\text{INPUT ANGLE}}{\text{DISPLAY ANGLE}}$$

$$SF/_{ELV} = \frac{90^\circ}{3.936^\circ} = 22.866$$

$$SF/_{AZ} = \frac{220^\circ}{9.632^\circ} = 22.841$$

(2) ALL DIMENSIONS ARE IN MILLIRADIANS (MR) EXCEPT AS NOTED.

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3.4.2.4.8 Numeric data location. As a minimum, each of the numerals shall be formed by one part of a seven segment pattern as shown in Figure 7. All numerals shall always be visible to at least one eye when viewed at the design eye position.

3.4.2.4.9 Brightness.

3.4.2.4.9.1 Displayed data brightness. The line brightness of all displayed data, symbols, and reticles shall be such that the images shall be clearly visible against 10,000 foot-lambert background brightness, and in the presence of direct sublight penetration in the cockpit. The minimum contrast ratio (C) under the conditions shall be 1.15. The contrast ratio (C) is defined as:

$$C = \frac{B_2 + B_1}{B_1}$$

Where: B_2 is the apparent brightness of the symbology as viewed thru the combiner.

B_1 is the apparent brightness of the background as viewed thru the combiner.

The projected image shall be clearly defined with no visible breaks or irregularities when viewed from a normal operating position in a darkened area or against a bright sky.

3.4.2.4.9.2 Brightness control range. The displayed data brightness shall be adjustable by controls and/or filtering to accommodate an ambient brightness range of 1×10^{-4} (1×10^{-5} desired) to 10,000 foot-lamberts on two ranges if necessary. For ambient brightness below 1.0 foot lamberts, AN/PVS-5 goggles may be worn by the pilot. Symbol brightness shall vary non-linearly with adjustment of the manual brightness control on the HUD display unit, such that the control sensitivity at a given contrast is approximately constant over the ambient brightness range.

3.4.2.4.9.3 Brightness uniformity. For a given brightness setting, the average line brightness of symbols shall not vary more than 25 percent over the FOV of the HUD. Brightness variation is defined as follows:

$$B \text{ variation} = \frac{B \text{ max} - B \text{ min}}{B \text{ max} + B \text{ min}}$$

Where: B max = maximum symbol line brightness
B min = minimum symbol line brightness

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3.4.2.4.9.4 Automatic Brightness Control (ABC). A light sensor and control system which automatically adjusts the brightness of the HUD and symbology in response to the outside ambient light level shall be incorporated in the HUD. A switch on the HUD front panel permits the pilot to enable or defeat the automatic control. The ABC shall operate within the brightness range of the symbols and for an ambient range of 100 to 6000 foot-lamberts.

3.4.2.5 Accuracy. Within the environmental conditions specified herein, the HUDS shall meet the following accuracies:

a. Fire control reticle positioning accuracies. Angular deviation in azimuth and elevation between the angular position of the center of the fire control reticle and the FCC commanded angular position shall not exceed 2 milliradians (mr), one sigma, in each axis. This shall apply within the angular limits shown in Figure 3. When the angular position of the fire control reticle is derived by the HUDS as a function of input laser range, the angular deviation in azimuth and elevation between the angular position of the center of the fire control reticle and the theoretical angular position (zero degree azimuth, ideal range-derived-elevation) shall not exceed $\pm$ 0.7 mr, one sigma, in azimuth and $\pm$ 2.7 mr, one sigma, in elevation. This shall apply within the elevation angular limits shown in Figure 3. All angular positions are with respect to the ADL.

b. Gunner's TSU LOS cueing reticle positioning accuracy. When not in TOW mode, the angular deviation in azimuth and elevation between the angular position of the center of the LOS cueing reticle and the TOW sight commanded angular position shall not exceed $\pm$ 4 mr, one sigma, in each axis. This shall apply within the angular limits shown in Figure 3. When in TOW prelaunch mode, the angular deviation shall be $\pm$ 3 mr, one sigma, in each axis. This shall apply within the angular limits shown in Figure II. When in TOW post launch mode, the angular deviation between the modified angular position (actual elevation and azimuth angles multiplied by 22.85) of center of the LOS cueing reticle and the TOW sight commanded angular position shall not exceed 4 degrees, one sigma, in each axis. This shall apply within the angular limits shown in Figure 12. All angular positions are with respect to the ADL.

c. TOW rectangular symbol accuracies. The azimuth angular width and elevation angular height for the TOW prelaunch rectangle symbol shall be as shown in Figure 9. The TOW postlaunch rectangle symbol shall be as shown in Figure 9.

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d. BRS positioning accuracy. The angular deviation in azimuth and elevation between the angular position of the center of the boresight reference symbol (BRS) and the ADL shall not exceed ± 0.7 mr, one sigma, in each axis.

e. SMR positioning accuracy. The angular deviation in azimuth and elevation between the angular position of the center of the SMR and the ADL shall not exceed ± 0.7 mr, one sigma, in each axis.

f. TOW coarse azimuth scale pointer accuracy. The azimuth angular deviation between the modified angular position (actual azimuth angle in degrees, measured with respect to the ADL, multiplied by 22.85) of the center of the scale pointer and the commanded coarse azimuth input shall not exceed ± 12 degrees.

3.4.2.6 Manual controls. The HUDS shall have the following manual controls on the front and side panels of the HUD.

3.4.2.6.1 Mode control. The mode control toggle switch on the front panel shall have the following positions:

a. NORMAL - All basic symbols are displayed as specified.

b. TEST - Initiates BIT operation and displays all symbols and numeric data segments.

c. STAD - SMR reticle displayed.

3.4.2.6.2 ABC/brightness. This control shall be a potentiometer adjustment, and shall adjust the operating point of the automatic brightness control subsystem. The pilot shall also be able to defeat ABC using a two position toggle switch.

3.4.2.6.3 Direct/indirect fire mode. The HUD shall incorporate a two position toggle switch which shall be a part of the HUD controls for use in helicopter fire control circuits.

3.4.2.6.4 Power control. The power control toggle switch on the front panel shall have the following positions:

a. OFF - Operating power interrupted.

b. STANDBY - Partial electrical power on, no symbols displayed.

c. ON - All necessary power is applied for HUD operations.

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3.4.2.7 Reflector/combiner plate. The combiner glass shall conform to MIL-O-13830.

3.4.2.8 Symbol color. To accommodate night operation without the AN/PVS-5 night vision goggles, it is desired that the symbols displayed on the HUD shall be red in color in accordance with MIL-L-25467.

3.4.2.9 Diagnostics. The HUDS shall have a self-test capability.

3.4.2.10 Built-in-test-equipment (BITE).

3.4.2.10.1 Fault detection/location. The HUDS shall have a built-in-test (BIT) capability to check the operational status of the equipment.

3.4.2.10.2 BITE output. The BITE provisions shall also provide for driving a remote indicator. This output shall be such that a signal of 28 vdc indicates a "go" condition, and an open signal indicates a "no-go" condition.

3.4.2.10.3 Malfunction indication. An equipment status indicator shall be provided for each LRU. The equipment status indicator shall be a "go"/"no-go" device and shall be such as to hold the last result if power is interrupted or removed, and shall be located on the surface of the LRU. Bite initiation and repetition shall be automatic and independent of pilot inputs.

3.4.2.11 Range data interface. Two sources of range data shall be accepted by the HUDS for the display of digital range data. These two sources shall be as follows:

a. Fire Control Computer Range

b. Laser Range

3.4.2.11.1 Range display precedence. Range shall be displayed from one of two sources, with the following priority, respectively:

a. FCC Range - Displayed when a valid FCC GOOD discrete is present at the HUD interface.

b. Laser Range - Displayed when a false FCC GOOD discrete is present at the HUDS interface and a valid laser range word in available at the HUDS interface.

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3.4.2.11.2 Depressed reticle solution. The fire control reticle for backup rocket (see Figure 2) shall be used as depressed reticle for rockets when the FCC fails (FCC GOOD discrete is false). When this occurs, the range data being used for range display on the HUD shall also be processed for depressing/elevating the symbol in accordance with an exponential curve fit for rockets.

3.4.3 Physical characteristics.

3.4.3.1 Dimensions. Dimensional requirements applicable to envelope size, mounting, and mating shall be in accordance with Drawings 11780001, 11780002, and 11780246.

3.4.3.2 Weight HUDS weight shall be 26 pounds maximum.

3.4.3.3 Panel lighting. The instrument shall be provided with integral 5-volt red lighting conforming to MIL-L-25467, except as follows:

a. Coating. Reflection reducing coating shall be in accordance with MIL-C-14806.

b. Reduced voltage operation. The lighting shall operate satisfactorily at reduced voltage. The relative brightness and color of the display shall be uniform and all markings shall be legible when the lighting is operated at 2.70 $\pm$ 0.02 Vdc and viewed from a distance of 28 to 30 inches.

3.4.4 Reliability.

3.4.4.1 System reliability. The HUDS shall have a mean-time-between-failure (MTBF) of no less than 1160 hours, under service conditions.

3.4.4.2 Service life. The HUDS shall have a minimum service life of 19,000 flight hours.

3.4.4.3 Preconditioning (burn-in). Each HUDS procured under this specification shall be subjected to a burn-in test as defined in Appendix B.

3.4.5 Environmental conditions (first article testing only). The HUDS shall operate and meet the performance requirements of 3.4.2 under any of the environmental conditions, or a reasonable combination of these conditions, as specified herein.

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3.4.5.1 Natural environment. The HUDS shall operate and withstand storage in the environments specified herein.

3.4.5.1.1 Altitude. Operating: 0-20,000 feet; nonoperating and storage: 0-50,000 feet.

3.4.5.1.2 Temperature. Operating nonoperating, and storage: -50° to +71°C.

3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 507.1, Procedure II. In Step 6, measurements shall be taken during the last five hours of the last test cycle.

3.4.5.1.4 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.1, Procedure I, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

3.4.5.1.5 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the HUDS shall not be required and failure criteria shall be limited to corrosion of finished and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

3.4.5.1.6 Fungus. No evidence of viable fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

3.4.5.1.7 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.1, Procedure I, except test to be limited to a maximum altitude of 20,000 feet.

3.4.5.1.8 Rain. Operation after being subjected to MIL-STD-810, Method 506.1, Procedure I. During each thirty minute rain period, the wind shall be maintained for fifteen minutes, commencing five minutes after the start of the period as specified in the procedures. The HUDS shall be operated during the last ten minutes of the thirty minute period.

3.4.5.2 Service environment. The HUDS shall withstand the service environment as specified in MIL-E-5400 for Class 2 equipment except as noted in Paragraphs 3.4.7.1.1 and 3.4.7.1.2.

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3.4.5.3 Induced environment.

3.4.5.3.1 Vibration. The HUDS shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Method 514.2, Procedure I, Part I, Table 514.2-III of MIL-STD-810; except that the frequency shall be limited to 500 Hz (reference Figure 18).

3.4.5.3.2 Shock. The HUDS shall meet the shock requirements of MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2 and Method 516.2, Procedure III.

3.4.5.3.2.1 Bench handling. The HUDS shall withstand shock environments encountered in maintenance and servicing.

3.4.5.3.3 Acceleration. The HUDS shall operate without degradation when exposed to acceleration levels up to 2g.

3.4.6 Transportability. The HUDS shall withstand vibration environments encountered during transporting as specified in MIL-STD-810, Method 514.2, Procedure X.

3.4.7 Soldering. All soldering shall be done in accordance with MIL-S-45743 or MIL-S-46844.

3.4.8 Electromagnetic radiation (first article testing only).

3.4.8.1 Electromagnetic compatibility. Each HUDS component as installed in the helicopter, shall not cause undesirable response, malfunction, or degraded performance to any other equipment/ subsystem installed in or associated with the helicopter; nor shall any component malfunction, exhibit degraded performance or undesirable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the helicopter are singly or collectively operated and performing their intended functions.

3.4.8.2 Electromagnetic interference. The HUDS shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE01, CE03, CE05, CS01, CS02, CS06, RE02, RS01, RS03, RS03.1, and RS02.

3.4.8.3 Circuit isolation. Signal and power potentials shall be electrically isolated from each other and the equipment chassis.

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3.4.9 Interchangeability (first article only). All components having the same manufacturer's part number shall be dimensionally and functionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto. Interchangeability must be maintained within the detail parts to the level of replaceable items.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.6 Workmanship. Workmanship shall be in accordance with electronic and mechanical weapon-build practices to ensure safety, proper and dependable operation, and service life. Particular attention shall be given to assure thoroughness of assembly, alignment of parts, electrical connections, cleanliness, tightness of screws and bolts, marking of parts, application of protective finish, and freedom from burrs and rough or sharp edges.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-E-5400.

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4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three HUDS selected at random by the Government representative from the first fifteen production HUDS.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table II.

4.4.2 Failure. Failure of any HUDS to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any HUDS to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE II. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Natural environment	3.4.5.1	4.6.3.1
2	Altitude	3.4.5.1.1	4.6.3.2
3	Temperature	3.4.5.1.2	4.6.3.3
4	Humidity	3.4.5.1.3	4.6.3.4
5	Sand and dust	3.4.5.1.4	4.6.3.5
6	Salt atmosphere	3.4.5.1.5	4.6.3.6
7	Fungus	3.4.5.1.6	4.6.3.7
8	Explosive atmosphere	3.4.5.1.7	4.6.3.8
9	Rain	3.4.5.1.8	4.6.3.9
10	Interchangeability	3.4.9	Applicable drawings, visual, MIL-I-8500, & MIL-STD-100
11	Service environment	3.4.5.2	4.6.3.10
12	Vibration	3.4.5.3.1	4.6.3.11.1
13	Shock	3.4.5.3.2	4.6.3.11.2
14	Bench handling	3.4.5.2.1	4.6.3.11.2.1
15	Acceleration	3.4.5.3.3	4.6.3.11.2
16	Transportability	3.4.6	4.6.3.11.3
17	Electromagnetic radiation	3.4.8	4.6.3.12

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TABLE III. Performance tests

Item	Characteristic	Requirement	Test Procedure
101	Electrical signal interface	3.4.1.1	Appendix A
102	Power requirements	3.4.1.2	4.6.2.1
103	Operation	3.4.2.1	4.6.2.1
104	Power control modes	3.4.2.2	4.6.2.1
105	Operational modes	3.4.2.3	4.6.2.2
106	Stadiametric mode	3.4.2.3.1	4.6.2.2
107	Normal mode	3.4.2.3.2	4.6.2.2
108	Boresight reference symbol	3.4.2.3.2.1	4.6.2.2
109	Normal data	3.4.2.3.2.2	4.6.2.2
110	Fire control symbol	3.4.2.3.2.3	4.6.2.2
111	TOW mode symbols	3.4.2.3.2.4	4.6.2.2
112	TSU LOS cueing symbol	3.4.2.3.2.5	4.6.2.2
113	TSU coarse azimuth symbols	3.4.2.3.2.6	4.6.2.2
114	Test mode	3.4.2.3.3	4.6.2.3
115	SMR characteristics	3.4.2.4.1	4.6.2.4
116	Boresight reference symbol	3.4.2.4.2	4.6.2.4
117	Numeric data	3.4.2.4.3	4.6.2.4
118	Fire control reticle	3.4.2.4.4	4.6.2.4
119	TOW mode symbols	3.4.2.4.5	4.6.2.4
120	Gunner's TSU LOS cueing symbol	3.4.2.4.6	4.6.2.4
121	TSU coarse azimuth symbols	3.4.2.4.7	4.6.2.4
122	Numeric data location	3.4.2.4.8	4.6.2.4
123	Displayed data brightness	3.4.2.4.9.1	4.6.2.4
124	Brightness control range	3.4.2.4.9.2	4.6.2.4
125	Brightness uniformity	3.4.2.4.9.3	4.6.2.4
126	Automatic brightness control (ABC)	3.4.2.4.9.4	4.6.2.4
127	Accuracy	3.4.2.5	4.6.2.5
128	ABC/brightness	3.4.2.6.2	4.6.2.6
129	Power control	3.4.2.6.4	4.6.2.6
130	Symbol color	3.4.2.8	4.6.2.8
131	Diagnostics	3.4.2.9	4.6.2.9
132	Fault detection/location	3.4.2.10.1	4.6.2.10
133	BITE output	3.4.2.10.2	4.6.2.10
134	Malfunction indication	3.4.2.10.3	4.6.2.10
135	Range data interface	3.4.2.11	4.6.2.11
136	Range display precedence	3.4.2.11.1	4.6.2.11
137	Depressed reticle solution	3.4.2.11.2	4.6.2.11
138	Burn-in	3.4.4.3	Appendix B

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TABLE IV. General tests

Item	Characteristic	Requirement	Test Procedure
301	Fabrication and General Specifications	3.1, 3.2, & 3.3	Applicable Drawings, Visual, and MIL-E-5400, Visual

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4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each HUDS in every lot shall be subjected to the tests in Table III and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Acceptance. Where any HUDS fails to meet any specified requirement, the defective HUDS shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One HUDS, as a control sample, shall be selected at random by the Government representative from each 100 HUDS produced and shall be subjected to the tests in Table IV.

4.5.2.2 Acceptance. Where any HUDS fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in Table V shall be used where specified to perform the required tests.

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TABLE V. Test equipment

Item/Nomenclature	Type		Mfr. & Model or Part No.
	Standard	Peculiar	
1. Digital Voltmeter	X		HP 3440A with Range Unit
2. Oscilloscope	X		Tektronix 547 with W Preamp
3. HUDS Test Set(M142)		X	11780600
4. Power Test Panel		X	Ohio Semitronics PTP 707
5. Optical Boresight Set Including:		X	
Boresight Bench			
Theodolite			Zeiss 70-24-15
Autocolimating Eye Piece			Zeiss 70-44-10
6. High Efficiency Photometric Telescope	X		Gamma Scientific 2020-31 SMR x/Microscope Eyepiece
7. Controller	X		Gamma Scientific DR1A
8. X/Y Plotter	X		HP 7045A-002
9. Ammeter	X		Commercial
10. Cable (W2)-		X	11780654
11. Cable (W3)- Part of M142		X	11780656
12. Cable (W4)- Test Set		X	11780657
13. Cable (W5)-		X	11780658
14. Flashlight	X		Commercial

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4.6.2 Performance tests.

4.6.2.1 Power control modes. Use items 3, 4, 5, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.2.

4.6.2.2 Operational modes. Use items 3, 4, 5, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.3 and 3.4.2.6.

4.6.2.3 Test mode. Use items 3, 4, 5, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.3.

4.6.2.4 Data and symbol characteristics. Use items 3, 4, 5, 6, 10, 11, 12, 13, and 14 of Table V to determine compliance with 3.4.2.4.

4.6.2.5 Accuracy. Use items 3, 4, 5, 6, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.5.

4.6.2.6 Manual controls. Use items 3, 4, 5, 6, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.6.

4.6.2.7 Reflector/combiner plate. The plate shall be examined in accordance with MIL-O-13830.

4.6.2.8 Symbol color. Use items 3, 4, 5, and 6 of Table V to determine compliance with 3.4.2.8.

4.6.2.9 Diagnostics. Use items 1, 2, 3, 4, 5, 7, 9, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.8.

4.6.2.10 Built-in-test-equipment (BITE). Use items 1, 2, 3, 4, 5, 7, 9, 10, 11, 12, and 13 of Table V to determine compliance with 3.4.2.9.

4.6.2.11 Range interface. Use items 3,4 5, 6, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.11.

4.6.3 Environmental tests.

4.6.3.1 Natural environment. Subject the system to the following environments:

4.6.3.2 Altitude. The system shall be subjected to the tests of Method 504.1 of MIL-STD-810 for Category 4 equipment. The system, nonoperating, shall be subjected to Method 500.1, Procedure I, with

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pressures simulating air transport at 50,000 feet altitude at a pressure of 2.85 inches of mercury (hg) for 1 hour. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.3 Temperature. The system shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810, except Step 4 temperature shall be 71°C. The system shall then be subjected to the test of Method 502.1, Procedure I, except maximum low temperature shall be -54°C. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.4 Humidity. The system shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810. In Step 6, measurements shall be taken during the last five hours of the last test cycle. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.5 Sand and dust. The system shall be subjected to test Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.6 Salt atmosphere. The system shall be subject to the test of Method 509.1 of MIL-STD-810. Operation of the subsystem shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.6.3.7 Fungus. The system shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces.

4.6.3.8 Explosive atmosphere. The system shall be subjected to test Method 511.1, Procedure I, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table III.

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4.6.3.9 Rain. The system shall be subjected to test Method 506.1, Procedure I, of MIL-STD-810. During each thirty minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedures. The system shall be operated during the last 10 minutes of the thirty minute period. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.10 Service environment.

4.6.3.10.1 Vibration. The system shall be subjected to the vibration test of MIL-E-5400, Figure 2 (Sheet 1 of 2), Curve III, except that the maximum double amplitude shall be 0.20 inch. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.10.2 Shock. The system shall be subjected to the shock test of Paragraph 3.2.2.4.6.1 of MIL-E-5400. The tests of Paragraph 3.2.2.4.6.2 of MIL-E-5400 for mounting hardware crash safety shall apply only to the units mounted inside the cockpit. At the conclusion of this test, subject the system to the performance test of TABLE III.

4.6.3.11 Induced environment.

4.6.3.11.1 Vibration. The system shall be subject to test Method 514.2, Procedure I, Part I, Table 514.2-III, of MIL-STD-810, except that the frequency shall be limited to 500 cycles (reference Fig 13). At the conclusion of this test, subject the system to the performance test of Table III.

4.6.3.11.2 Shock. The system shall be subject to:

a. Method 516.2 of MIL-STD-810, Procedure I, Figure 516.2-2 for flight vehicle equipment. The test "g" value shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter. The pulse test level and duration are defined in Paragraph 3.2.24.6.1 of MIL-E-5400.

b. Method 516.2, Procedure III for mounting base crash safety. The pulse test level and duration are defined in Paragraph 3.2.24.6.2 of MIL-E-5400.

At the conclusion of this test, subject the system to the performance test of Table III.

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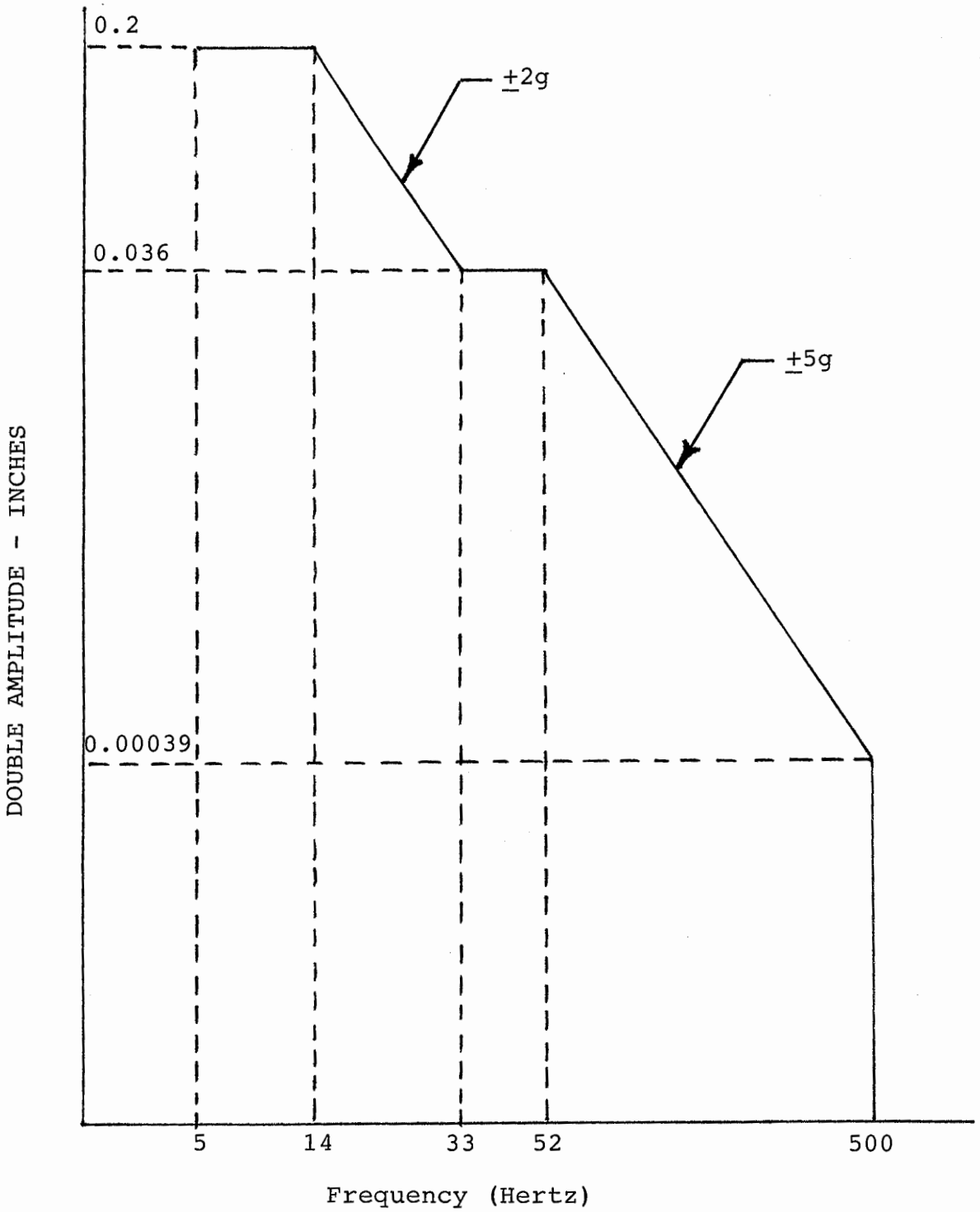


FIGURE 13. Vibration amplitude for first article test

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4.6.3.11.2.1 Bench handling. The system shall be subject to Method 516.2, Procedure V, of MIL-STD-810. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.11.3 Transportability. The system shall be subject to the test of Method 514.2, Procedure X, of MIL-STD-810. At the conclusion of this test, subject the system to the performance tests of Table III.

4.6.3.12 Electromagnetic radiation. The subsystem shall be installed in laboratory test fixtures in a manner that will simulate actual service useage and tested to MIL-STD-461 for the requirements of 3.4.8.

5. PACKAGING.

5.1 Preservation, packing, packaging and marking for the sight, helicopter head-up subsystem: M76, Dwg. 11780000 shall be in accordance with the respective packaging data sheets listed below at the level of protection specified in the contract (see 6.2B):

a. P11780001, Display Unit, Sight, Helicopter Head-Up Subsystem: M76.

b. P11780002, Processor, Signal, Sight, Helicopter, Head-Up Subsystem: M76.

c. P11780246, Mount, Head-Up Display: M76.

5.2 Commercial packaging shall be in accordance with ASTM-D3951.

6. NOTES.

6.1 Intended use. This equipment is intended for use in the AH-1S (modernized) helicopters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging Data Sheet (see 5.1).

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c. Selection of applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

e. First article inspection requirements.

6.3 Previous revision of MIL-S-70442 (AR) was for the US Army Armament Research and Development Center internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270-A049)

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APPENDIX A

HUDS SIGNAL INTERFACE

REQUIREMENTS

Detail characteristics. The HUD communicates with the Fire Control Computer (FCC), the TOW Missile System (TMS), the AN/APM 209 radar altimeter, Airborne Laser Tracker (ALT), and the Laser Range Finder (LRF). Data transmission from the FCC and from the LRF is by means of dedicated serial-data busses. Signals from the TMS and the radar altimeter are in the form of parallel discrete and analog signals.

HUD-FCC interface. The FCC transmits data to HUD by means of dedicated serial digital bus. The line consists of a twisted, shielded wire pair which carries the data and its logical complement. The wiring configuration is shown in Figure 14. A timing diagram illustrating typical signal timing is shown in Figure 15.

Bus characteristics. The data line-pair between the FCC and the HUD consists of a twisted shielded wire pair with a characteristic impedance of 71 ohms, ±10 percent. The shield on this pair is grounded in accordance with good EMI suppression practices. Data is transmitted and received by differential 3-state logic transmitters and receivers. The signal levels used shall be TTL compatible.

Data format. The data is sent asynchronously at a normal 9600 bit/second rate. Each data byte forms a part of an 11-bit transmission period consisting of a start bit, 8-bit data byte, stop bit, and parity bit, as shown in the timing diagrams of Figure 15. Transmitted parity, including parity bit, is odd. Two such bytes sent sequentially make up a 16-bit word.

Message format. The information sent in the data words are in the sequence shown in Table I. The FCC sends all six words, two bytes per word, each data transmission.

HUD-LRF interface. The LRF transmits data to the HUDS on a serial digital data bus like that for the FCC. The wiring configuration is as shown in Figure 14. In the event of FCC failure, data transmission is to be received directly from the LRF with signal timing as shown in Figure 16.

HUD-TMS interface. The HUDS replaces the Pilot's Steering Indicator (PSI) in the TMS and so receives all the signal information which the PSI would normally receive.

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HUD-radar altitude interface. The HUDS receives low and high limit discretes directly from the radar altimeter. Radar altitude comes to the HUDS as a digital work in the data from the FCC. In case of FCC malfunction, radar altitude information may be absent. The HUDS continue to generate high and low warning symbols normally in response to these discretes in that case.

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TABLE II. Data Word Structure.

Parameter	Displayed Resolution	Parameter Resolution	Parameter Range	Number of Bits Required	Byte Positions in Data Transfer Sequence
Range Smoothed	10m	1m	0.2 to 9.99 km	14 Bits	1 and 2
Radar Altitude	5 ft	1 ft	0 to 5000 ft	13 Bits	3 and 4
Magnetic Heading	1°	1°	0 to 359°	9 Bits	5 and 6
Torque	1%	1%	0 to 145%	10 Bits	7 and 8
FC Reticle AZ	0.3582 mr	0.1 mr	±157 mr	11 Bits + Sign (Extended sign two's complement)	9 and 10
FC Reticle EL	0.3582 mr	0.1 mr	+131 mr -184 mr	11 Bits + Sign (Extended sign two's complement)	11 and 12
TSU AZ	0.3582 mr	0.1 mr	±110°	15 Bits + Sign (Extended sign two's complement)	13 and 14
TSU EL	0.3582 mr	0.1 mr	+30° -60°	14 Bits + Sign (Extended sign two's complement)	15 and 16

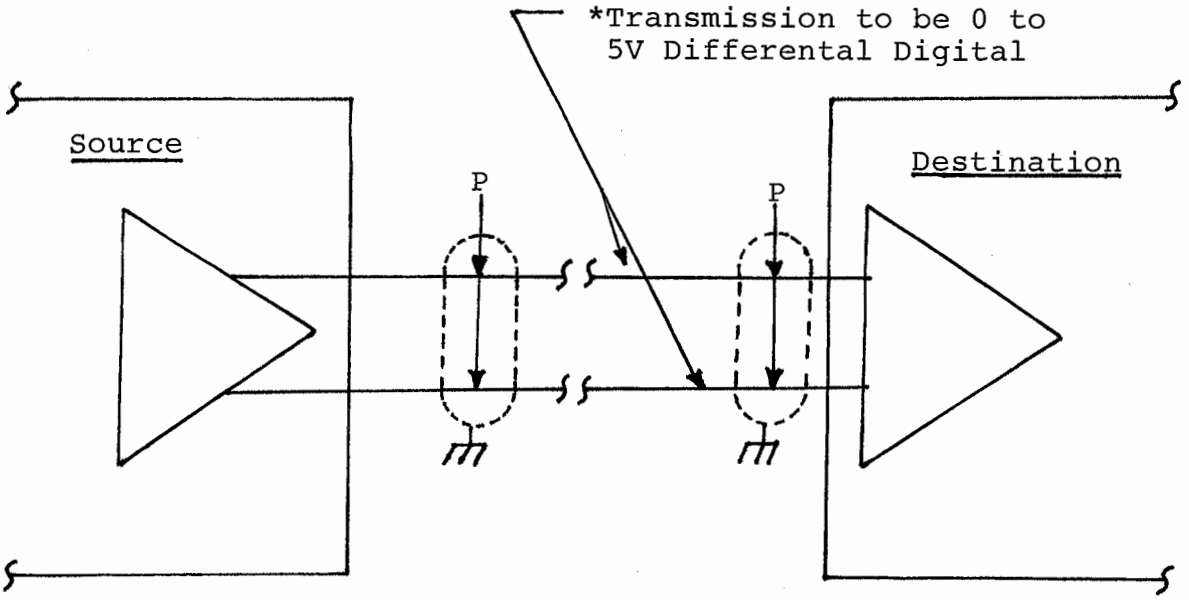
DATA BLOCK TRANSFER

byte 1	byte 2	byte 3	byte 4	byte 5	byte 6	byte 7	byte 8	byte 9	byte 10	byte 11	byte 12	byte 13	byte 14	byte 15	byte 16

Start of Data Block

End of Data Block

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Output Capability

Fairchild 9614 or Equivalent

Baud Rate = 9600

Input Capability

Fairchild 9615 or Equivalent

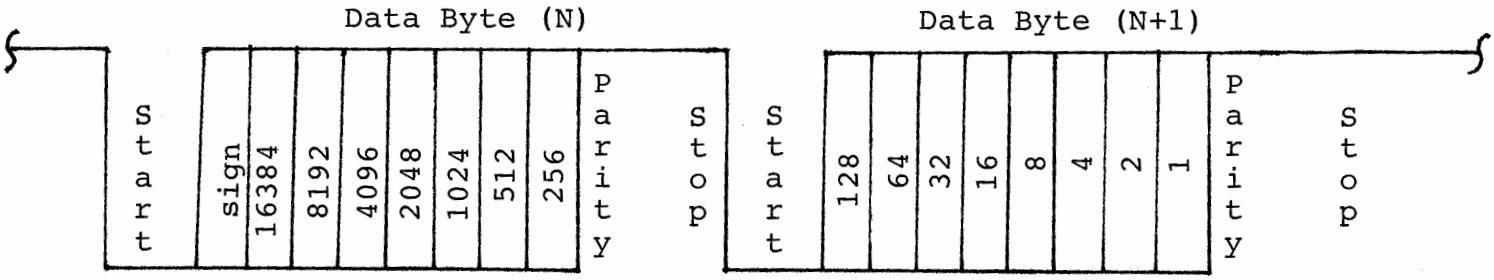
Baud Rate = 9600

*

	Lines	
	True	False
Logic "1"	+2.8 To +5.0 V	-0.5 To +0.8 V
Logic "0"	-0.5 To +0.8 V	+2.8 To +5.0 V

Figure 14. Interface Bus.

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Where: N = 1, 3, 5, 7, 9, 11
N+1 = 2, 4, 6, 8, 10, 12

Note: All Data is to be 16 BIT 2's Complement Binary Data Scaled from the LSB (1) as defined in Parameter Resolution Column of Table I.

Each byte is 11 Bits.

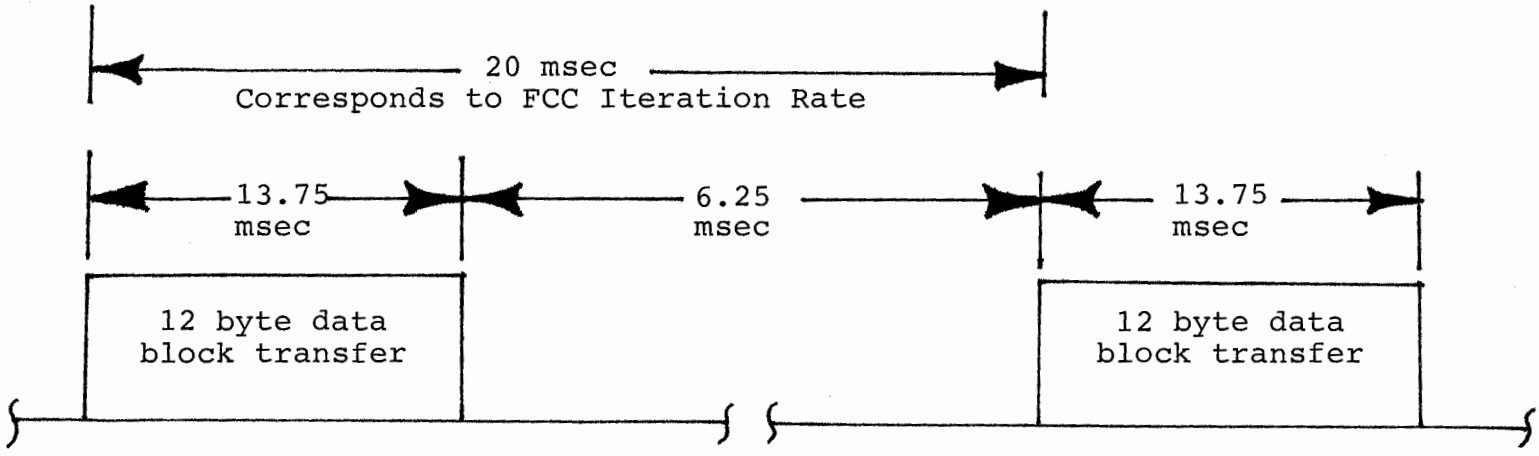
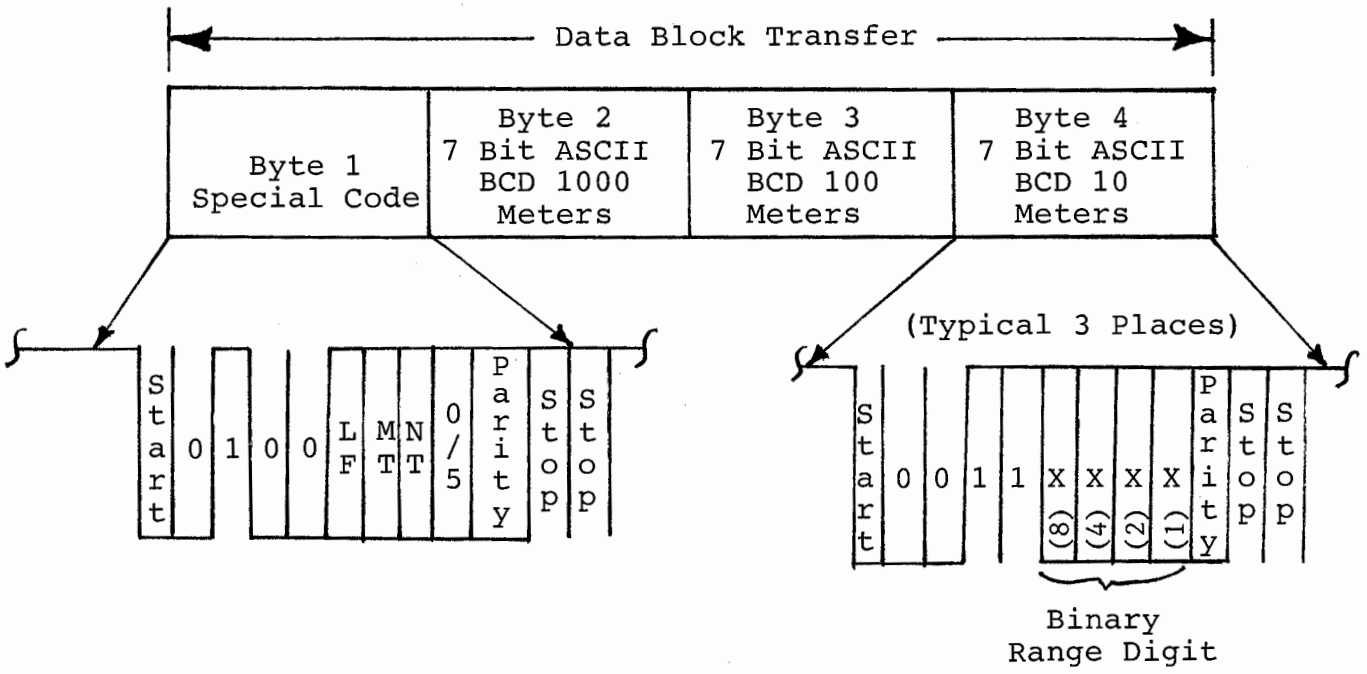
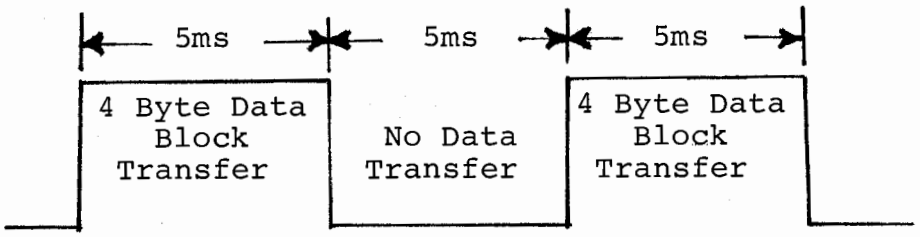


Figure 15. Data Transmission from FCC to HUD.

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Data Transfer Timing



Special Code Defined

LF = Laser Fired when True
MT = Multi-Target when True
NT = No Target Return when True
0/5 = 5 Meter Incr.

Odd True Parity

Figure 16. Laser Range Finder Data Format.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 20

NOMENCLATURE	FCC GOOD
TYPE	28 VDC
CONDUCTOR TYPE AND GAUGE	SINGLE UNSHIELDED
ORIGIN	FIRE CONTROL COMPUTER
DESTINATION	SEE REMARKS
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	FCC, RELAY
SOURCE IMPEDANCE	N/A
MAX CURRENT	0.25 AMPERE TOTAL
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC (FCC GOOD)
FALSE STATE VOLTAGE	OPEN CIRCUIT (FCC FAIL)

REMARKS DESTINATIONS AND DESTINATION IMPEDANCES ARE:

1. IFCU, RELAY COIL, 186 OHMS OR GREATER, WITH SURFACE DEVICE TO AVOID TURN-OFF TRANSIENTS.
2. PILOT'S CAUTION PANEL, RELAY COIL, 550 OHMS OR GREATER, WITH SUITABLE DEVICE TO AVOID TURN-OFF TRANSIENTS.
3. HUDS, $\geq$6.6 K OHMS.
4. ROCKET MANAGEMENT SUBSYSTEM, $\geq$2K OHMS.

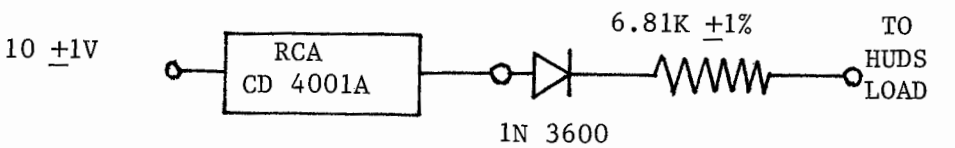
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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 32

NOMENCLATURE	DESCENT COMMAND
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$20 K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	7K OHMS (SEE REMARKS)
MAX CURRENT	1 MA
MAX VOLTAGE	11V
TRUE STATE VOLTAGE	+10 $\pm$ 1V (DESCEND)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

10 VDC CAUSES HUD TO DISPLAY DESCEND COMMAND SYMBOL WHEN IN TOW MODE OF FIRE.



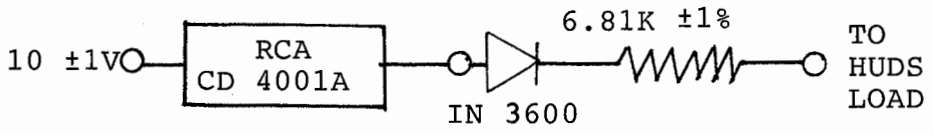
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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 33

NOMENCLATURE	ASCEND COMMAND
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$20 K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	7K OHMS (SEE REMARKS)
MAX CURRENT	1 MA
MAX VOLTAGE	11V
TRUE STATE VOLTAGE	+10 $\pm$ 1V (ASCEND)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

10 VDC CAUSES HUD TO DISPLAY ASCEND COMMAND SYMBOL WHEN IN TOW MODE OF FIRE.



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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 34

NOMENCLATURE	ATTACK MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$<$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC(ATTACK)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

28 VDC CAUSED HUD TO DISPLAY TOW MODE PRELAUNCH ATTACK SYMBOLS.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 35

NOMENCLATURE	READY MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geqslant$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$<$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (READY)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

28 VDC CAUSED HUD TO DISPLAY TOW MODE PRELAUNCH ATTACK SYMBOLS.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 36

NOMENCLATURE	FIRE/MANEUVER MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$<$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (MANEUVER)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

AFTER THE TOW MISSILE HAS BEEN FIRED, A 28 VDC DISCRETE CAUSES THE HUD TO DISPLAY TOW MODE POSTLAUNCH SYMBOL.

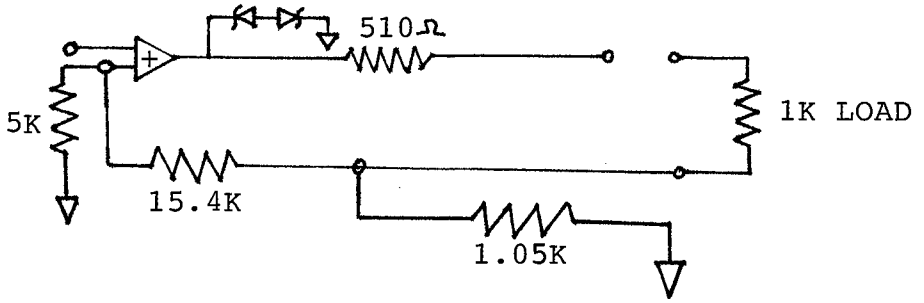
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Check the source to verify that this is the current version before use.

DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 37

NOMENCLATURE	TSU AZ ANGLE
TYPE	VARIABLE DC (CURRENT DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	(CURRENT SOURCE)
VOLTAGE RANGE	N/A
SCALE FACTOR	SEE REMARKS
PARAMETER RANGE	SEE REMARKS
PARAMETER ACCURACY	SEE REMARKS
PHASING	+ CURRENT IS LOS TO THE RIGHT
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	± 2.2 ma (ALSO SEE REMARKS)
MAX VOLTAGE	± 2.2 VOLTS (ALSO SEE REMARKS)
MAX SLEW RATE	SEE REMARKS

REMARKS

1. PRELAUNCH AND NON-TOW MODE CUEING SIGNAL CHARACTERISTICS: ±0.4 MA/DEGREE ±4% OVER ±3.75 DEGREES AT ±250 MR/SEC MAX.
2. POSTLAUNCH SIGNAL CHARACTERISTICS (TOW MODE ONLY): ±0.0136 MA/DEGREE ±5% OVER ±110 DEGREES AT ±1.7 RAD/SEC MAX.
3.



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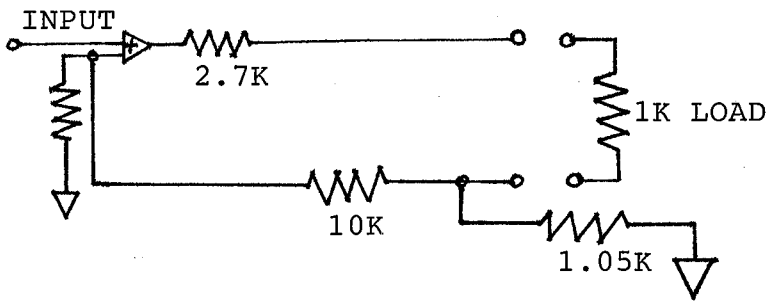
DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 38

NOMENCLATURE	TSU EL ANGLE
TYPE	VARIABLE DC (CURRENT DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	(CURRENT SOURCE)
VOLTAGE RANGE	N/A
SCALE FACTOR	SEE REMARKS
PARAMETER RANGE	SEE REMARKS
PARAMETER ACCURACY	SEE REMARKS
PHASING	+ CURRENT IS LOS UP
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	± 2.2 ma (ALSO SEE REMARKS)
MAX VOLTAGE	± 2.2 VOLTS (ALSO SEE REMARKS)
MAX SLEW RATE	SEE REMARKS

REMARKS

THE NORMAL OPERATING RANGE FOR THIS SIGNAL IS:

1. PRELAUNCH AND NON-TOW MODE CUEING SIGNAL CHARACTERISTICS: ±0.1 MA/DEGREE ±4% OVER ±7.5 to -15 DEGREES AT ±250 MR/SEC MAX.
2. POSTLAUNCH SIGNAL CHARACTERISTICS (TOW MODE ONLY): ±0.027 MA/DEGREE ±10% OVER ±30 to -60 DEGREES AT ±1.7 RAD/SEC MAX.
3.



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ANALOG SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 39

NOMENCLATURE	AZ COARSE POINTER
TYPE	VARIABLE DC (VOLTAGE DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS $\pm$10%
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$\geq$5.76K OHMS
VOLTAGE RANGE	$\pm$1.5 VDC
SCALE FACTOR	.0136 v/DEG
PARAMETER RANGE	$\pm 110^{o}$
PARAMETER ACCURACY	$\pm$10% OF ACTUAL
PHASING	+ VOLTAGE IS LOS TO THE RIGHT
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	$\pm$2.2 ma
MAX VOLTAGE	$\pm$2.2 VDC

REMARKS

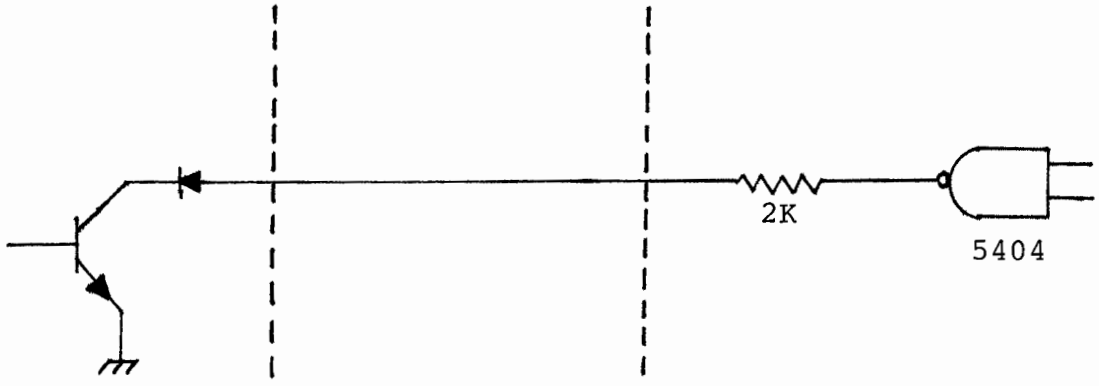
This symbol is used to position the TSU AZ LOS coarse symbol on the HUD when the gunner's LOS symbol is being used on the HUD.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 40

NOMENCLATURE	LOW ALTITUDE LIMIT
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	(SEE REMARKS)
SOURCE	RADAR ALTIMETER - APN-209
SOURCE IMPEDANCE	N/A
MAX CURRENT	10 ma
MAX VOLTAGE	5.5 VDC
TRUE STATE VOLTAGE	TRANSISTOR SWITCH CLOSED (ALTITUDE BELOW LIMIT)
FALSE STATE VOLTAGE	TRANSISTOR SWITCH OPEN (ALTITUDE ABOVE LIMIT)

REMARKS



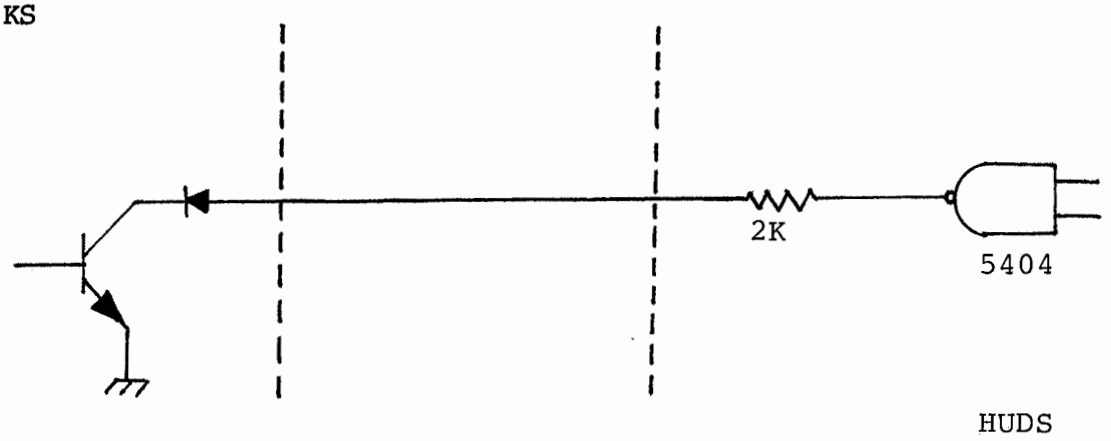
Radar Altimeter

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 41

NOMENCLATURE	HIGH ALTITUDE LIMIT
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	(SEE REMARKS)
SOURCE	RADAR ALTIMETER - APN-209
SOURCE IMPEDANCE	N/A
MAX CURRENT	10 ma
MAX VOLTAGE	5.5 VDC
TRUE STATE VOLTAGE	TRANSISTOR SWITCH CLOSED (ALTITUDE BELOW LIMIT)
FALSE STATE VOLTAGE	TRANSISTOR SWITCH OPEN (ALTITUDE ABOVE LIMIT)

REMARKS



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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 42

NOMENCLATURE	FIXED FORWARD GUN MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT POWER
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$ 6.6K OHMS
SOURCE	PILOT'S ACTION SWITCH
SOURCE IMPEDANCE	$\leq$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC
FALSE STATE VOLTAGE	OPEN

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 43

NOMENCLATURE	SPARE DISCRETE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geqslant$6.6K OHMS
SOURCE	TBD
SOURCE IMPEDANCE	$<$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC
FALSE STATE VOLTAGE	OPEN

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 44

NOMENCLATURE	DIRECT/INDIRECT ROCKET MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	FCC
DESTINATION IMPEDANCE	10K OHMS ±20%
SOURCE	HUD (CONTROLS)
SOURCE IMPEDANCE	N/A
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (INDIRECT)
FALSE STATE VOLTAGE	OPEN (DIRECT)

REMARKS

Used by FCC to compute fire control reticle location for direct and indirect fire mode.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 46

NOMENCLATURE	SPARE DISCRETE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geqslant$6.6K OHMS
SOURCE	TBD
SOURCE IMPEDANCE	$<$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (DISPLAY SYMBOL)
FALSE STATE VOLTAGE	OPEN

REMARKS

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DIGITAL SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 48

NOMENCLATURE	HUD DATA
TYPE	BINARY 2's COMPLEMENT
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	FCC
SOURCE IMPEDANCE	$\leq$100 OHMS
ADDRESS	N/A
NO. OF BITS/WORDS	8 BITS/1 PARITY/1 START/ 1 STOP=11 BITS
FIRST TRANSMITTED BIT	START BIT
SCALE FACTOR	N/A
LSB VALUE (RESOLUTION)	SEE TABLE I, FIGURES 2 and 3
PARAMETER RANGE	SEE TABLE I, FIGURES 2 and 3
PARAMETER ACCURACY	SEE TABLE I, FIGURES 2 and 3
TRANSMISSION RATE	9600 BAUD
ITERATION RATE	50 Hz
TRUE STATE VOLTAGE	TTL COMPATIBLE
FALSE STATE VOLTAGE	TTL COMPATIBLE

REMARKS

This signal must be compatible with Fairchild 9615, or equivalent, differential TTL receiver.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 49

NOMENCLATURE	LEFT MISSILE FIRING
TYPE	DC
CONDUCTOR TYPE AND GAUGE	TWO SINGLE-CONDUCTORS
ORIGIN	
DESTINATION	HUD
DESTINATION IMPEDANCE	N/A
SOURCE	TMS
SOURCE IMPEDANCE	>200 OHM
MAX CURRENT	50 ma
MAX VOLTAGE	N/A
TRUE STATE VOLTAGE	GROUND $\pm$1V
FALSE STATE VOLTAGE	OPEN

REMARKS

A left missile is to be fired from the aircraft. The selection of a missile will ground the left signal conductor depending on its launch side position.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 50

NOMENCLATURE	RIGHT MISSILE FIRING
TYPE	DC
CONDUCTOR TYPE AND GAUGE	TWO SINGLE-CONDUCTORS
ORIGIN	
DESTINATION	HUD
DESTINATION IMPEDANCE	N/A
SOURCE	TMS
SOURCE IMPEDANCE	$\geq$200 OHM
MAX CURRENT	50 ma
MAX VOLTAGE	N/A
TRUE STATE VOLTAGE	GROUND $\pm$1V
FALSE STATE VOLTAGE	OPEN

REMARKS

A right missile is to be fired from the aircraft. The selection of a missile will ground the right signal conductor depending on its launch side position.

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DIGITAL SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 52

NOMENCLATURE	RAW LRF DATA
TYPE	ASCII 7 BIT
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/TSU-LRF
DESTINATION	HUDS
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	TOW/TSU-LRF
SOURCE IMPEDANCE	$<$100 OHMS
ADDRESS	N/A
NO. OF BITS/WORDS	8 BITS/1 PARITY/1 START/ 2 STOP=12 BITS
FIRST TRANSMITTED BIT	START BIT
SCALE FACTOR	N/A
LSB VALUE (RESOLUTION)	SEE FIGURE 4
PARAMETER RANGE	SEE FIGURE 4
PARAMETER ACCURACY	N/A
TRANSMISSION RATE	9600 BAUD
ITERATION RATE	SEE FIGURE 4
TRUE STATE VOLTAGE	TTL COMPATIBLE (SEE FIGURE 2)
FALSE STATE VOLTAGE	TTL COMPATIBLE (SEE FIGURE 2)

REMARKS

This signal must be compatible with Fairchild 9615, or equivalent, differential TTL receiver. This signal will be continuously sent to the HUD. The HUD will display raw laser range data when "FCC GOOD" (discrete #20) false state voltage occurs.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 98

NOMENCLATURE	HUD GO/NO-GO
TYPE	BIT DISCRETE
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	
DESTINATION	AIRCRAFT CAUTION PANEL
DESTINATION IMPEDANCE	>560 OHMS
SOURCE	HUDS
SOURCE IMPEDANCE	10 OHM
MAX CURRENT	60 ma
MAX VOLTAGE	28 VDC (MIL-STD-704)
TRUE STATE VOLTAGE	+28 VDC = GOOD
FALSE STATE VOLTAGE	OPEN = NO/GO

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 118

NOMENCLATURE	DISPLAYS BRIGHTNESS MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/HUD
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	PILOT STATION CONTROLS
SOURCE IMPEDANCE	$\leq$10 OHM
MAX CURRENT	20 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC (LOW LIGHT - DIM)
FALSE STATE VOLTAGE	OPEN (NORMAL VIEWING - BRIGHT)

REMARKS

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APPENDIX B

BURN IN REQUIREMENTS

Burn in is to be applied to all deliverable systems or LRU's.

Test Environments

Environmental exposure for the HUDS during burn in is Level F of MIL-STD-781B. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5°C per minute. Figure 18 illustrates one cycle of the environmental profile described below.

Temperature	-54°C to +71°C (-65°F to 160°F)
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling NOTE: Voltage Cycling will occur on the primary HUDS power (28VDC) only.	A. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. B. Momentary off. C. 100% of nominal for next 1/3 of "ON" time. D. Momentary off. E. 90% of nominal for last 1/3 of "ON" time.

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Number of Cycles

The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

Duty Cycle Information

During equipment ON time, mode selection time shall be apportioned as follows:

Test	95% ± 5%

During operation the display and BIT Indicators shall be monitored.

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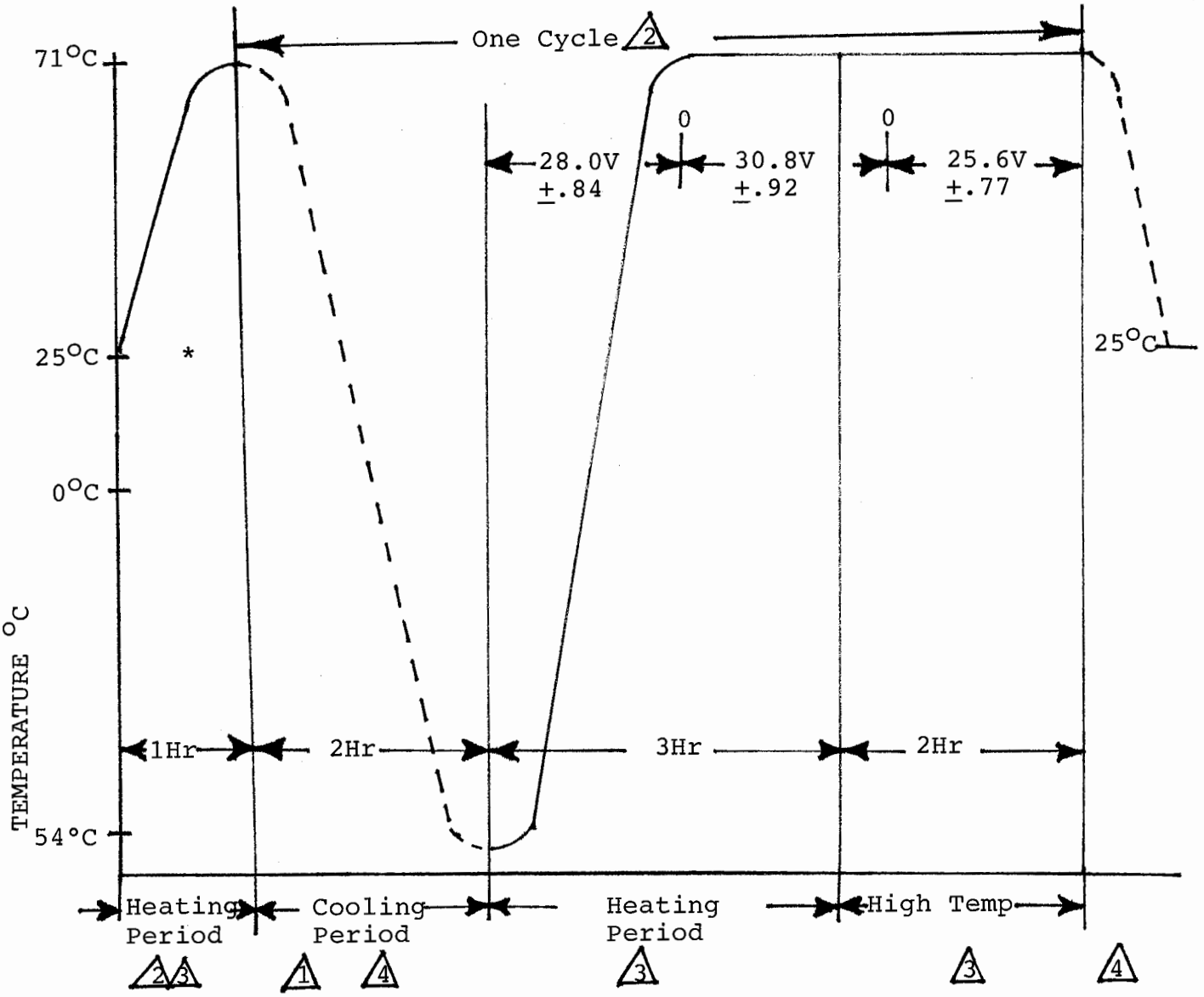


FIGURE 3-1

Typical Temperature Cycle-Electronic and Avionic Equipment
● Stabilized Temperature
0 Momentary Off
---- Equipment Off
——— Equipment On
* To Use When Chamber is to be Dehumidified to Protect Optics

FIGURE 18. Burn In Cycle.

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NOTES:

1. During the cooling period, the chamber temperature shall not be lower than -60°C.

2. One temperature cycle shall start with the equipment at high temperature and off (cooled and stabilized) at low temperature for two hours. During the cycle, the equipment shall operate in accordance with its normal duty cycle. The equipment will be on for five hours, two of which will be at high temperature. The chamber will be dehumidified when operating the HUD to protect the optics.

3. Equipment ON; input voltage (28 VDC) at 100%, 100%, and 90%, respectively, during 1/3 of the total time of heating period plus high temperature period; momentarily off between input voltage changes.

4. Equipment OFF.

5. Vibration - 2.2G $\pm$ 10% peak acceleration value at any non-resonant frequency between 20 and 60 Hertz (Hz) measured at the mounting points of the equipment. The vibration shall be at least 10 minutes during each hour of operating time.

6. Transistion rate between temperature levels shall average 5°C per minute.

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